

Geographic coverage debt of frozen Earth-observation foundation models: an encoder-robustness reliability diagnostic and a conditional target-recalibration remedy

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Abstract—Frozen geospatial foundation models (GFM) are increasingly deployed as off-the-shelf encoders for land-cover and local-climate-zone (LCZ) monitoring, yet practitioners lack guidance on which frozen encoder to trust after a regional move, or on how to restore a reliability guarantee once shift breaks it. We supply both through a distribution-free reliability audit (split-conformal, finite-sample guarantees) of frozen GFMs under real cross-region shift (thirteen encoders on GEO-Bench BigEarthNet-S2 for the FNR analysis; five on EuroSAT-MS (1) and non-European m-so2sat; the EuroSAT shift is partly a within-Europe label-prior shift, ruled out by a class-balanced control as the sole driver). This audit is packaged as a reusable pre-deployment reliability report-card protocol (§IV-B): for any frozen GFM, it returns four practitioner-facing numbers before deployment—the in-distribution FNR guarantee, the geographic debt that guarantee incurs under shift, the labeled-data cost of repairing that debt with a target-region slice, and the worst-class coverage a marginal guarantee alone would hide. Every frozen encoder meets a multi-label false-negative-rate (FNR) risk-control guarantee in-distribution, yet all thirteen incur a real geographic debt under shift (FNR $2.6\text{--}4.7\times$ nominal), bought back with larger sets, not a better encoder: FNR robustness rises with set size (Spearman $\rho = -0.66$, $p=0.017$), the paper’s one adequately powered cross-encoder result. A deterministic boundary bounds when the audit is worth running at all: marginal single-label sets collapse to singletons above accuracy ≈ 0.80 , where the conformal layer adds nothing beyond accuracy. The textbook unlabeled fix does not close the debt—covariate-shift-weighted conformal recovers only part of it, its effective sample size collapsing to $0.1\text{--}1.6\%$ of the calibration set—so only a labeled target-region slice (e.g. 30% of target data) reliably restores the guarantee, at the cost of large but still-informative restored sets (22–24 of ~ 27 active labels); this repair is expected by construction, a verification rather than an independent discovery. The cross-encoder question is weaker: the least accurate encoder (Prithvi) is the most FNR-robust, but the rank correlation with accuracy is only a weak, non-significant positive at $n=13$ ($\rho = +0.32$, $p=0.28$)—directional context, not a proven inversion, since a five-encoder pilot’s stronger correlation regressed toward zero on expansion. As a single-encoder, single-class existence proof, restoring the *marginal* coverage guarantee can also hide a per-class collapse invisible to accuracy and ECE. The audit is diagnostic, reusing existing conformal estimators; the preregistration outcome and exploratory-analysis disclosures are in §V.

Index Terms—Earth observation, geospatial foundation mod-

els, conformal prediction, distribution shift, uncertainty quantification, reliability, calibration.

I. INTRODUCTION

GEOSPATIAL foundation models are pretrained once and reused frozen as feature extractors for many downstream tasks. Beyond the encoders audited here (thirteen on BigEarthNet, five on EuroSAT and m-so2sat), frozen GFM backbones now span single-sensor optical pretraining (SatMAE (2), Scale-MAE (3), SeCo (4), GFM (5)), multi-sensor and multi-modal pretraining (SkySense (6), AnySat (7), TerraFM (8), Galileo (9)), and pixel-timeseries pretraining (Presto (10)); none of these families is conformally audited under geographic shift, and our roster (Methods) is illustrative of the diagnostic rather than a leaderboard survey of this rapidly growing space. Their accuracy under geographic shift has been studied (11, 12). Calibration itself is not entirely unstudied: SHRUG-FM (11) evaluates SSL4EO ViT-S/16 (MoCo/MAE/DINO) checkpoints on burn-scar segmentation (ExEBench) and reports expected calibration error (ECE $0.0248\text{--}0.0328$), but it has no conformal-prediction arm, no spatial-holdout cross-validation, and does not evaluate Clay or Prithvi. Uncertainty quantification for deep networks more broadly spans Bayesian, ensemble, and distribution-free approaches (13); conformal prediction is the distribution-free branch we build on, and it itself has entered Earth observation: Singh et al. (14) bring split and (class-conditional) Mondrian conformal prediction to EO pipelines (Google Earth Engine), GeoConformal (15) adds geographically-weighted conformal intervals for *spatial regression*, and Mao et al. (16) develop model-free conformal prediction for spatially indexed data under local exchangeability; none of the three audits frozen foundation-model encoders, nor coverage loss under an explicit geographic train/deploy shift. Weighted conformal methods for covariate shift (17) are the natural theoretical remedy but require density-ratio estimates we do not assume here. Closest in spirit, Fillioux et al. (18) ask whether frozen vision foundation models are good conformal predictors, but on in-distribution image classification with no geographic-shift audit; and Aolaritei et al. (19) develop Lévy–Prokhorov theory for conformal robustness to distribution shift—complementary theoretical machinery that they do not instantiate on Earth observation. Our geographic-shift framing is also distinct from the broader unsupervised-domain-adaptation literature for remote sensing classification

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(20, 21), which typically retrains or adapts the feature extractor itself against the target domain; we instead ask whether a *frozen*, never-retrained encoder’s conformal guarantee survives a regional shift, and how to repair it with only a labeled target-region calibration slice, not further training. Consequently, *conformal coverage* reliability under shift—whether a conformal predictor’s guarantee (22, 23) on frozen classification GFM such as Clay and Prithvi survives a move to a new region—has received far less attention. We ask a sharp, falsifiable question: under a real geographic shift, does a conformal audit of a frozen GFM tell a practitioner anything that shifted accuracy and calibration do not? We find that it does, in two regimes, and we characterize precisely where it does not. Our primary contribution is a reusable pre-deployment reliability report-card protocol (§IV-B, Table XI): apply one fixed audit to any frozen GFM and read off four practitioner-facing numbers before deployment—its in-distribution FNR guarantee, the geographic debt that guarantee incurs under shift, the labeled-data cost of repairing that debt with a target-region slice, and the worst-class coverage a marginal guarantee alone would hide. We introduce no new conformal estimator; the audited findings below are what make each report-card entry non-obvious and worth running the protocol for. The protocol’s first evidence axis is a within-encoder reliability diagnostic that needs no cross-encoder statistics: under a conformal-risk-control (24) bound on the example-averaged false-negative rate (FNR) on GEO-Bench BigEarthNet-S2, *every one* of thirteen frozen encoders meets the FNR guarantee in-distribution yet incurs a real geographic debt under shift (0.130–0.234, $2.6\text{--}4.7\times$ nominal), which a labeled target-region slice repairs (0.002–0.036)—a repair that is expected by construction, since the target-calibrated guarantee is a verification, not an independent discovery. This debt is bought back with larger prediction sets, and that trade is the one adequately powered cross-encoder result in the paper: FNR robustness rises with prediction-set size (Spearman $\rho = -0.66$ at $\alpha=0.10$, $p=0.017$; Table I), an operating-point choice no accuracy scalar can name. The protocol’s second evidence axis is the singleton-degeneracy boundary, a deterministic property rather than a correlation: the marginal single-label set collapses to a singleton above accuracy ≈ 0.80 , so EuroSAT at full data is an explicit *negative control* bounding the audit’s validity regime (So2Sat/BigEarthNet sit below it)—telling a practitioner exactly when a marginal audit of a frozen GFM is worth running at all. The protocol’s third evidence axis is a remedy-failure result, and it is what fixes the report card’s target-slice repair cost as necessary rather than optional: the textbook unlabeled fix for covariate shift does not close the debt. Covariate-shift-weighted conformal (17) recovers only part of the coverage loss because its weights’ effective sample size collapses to 0.1–1.6% of the calibration set—the density ratios are too heavy-tailed to be usable—so only a small labeled target-region slice reliably restores the guarantee; oracle and deployable label-shift reweighting fail outright (worsening coverage for three of four encoders, ruling out a pure class-prior effect and pointing instead to a within-class conditional shift); and spatial-Mondrian conditioning adds nothing measurable over non-spatial target calibration,

so the operative axis the protocol reports is access to labeled target-region data, not a new estimator (scope and estimator provenance are stated in §V). Only after these three do we report the weaker cross-encoder accuracy question, as directional context rather than a claim: accuracy is a weak and unreliable guide to *which* encoder is most robust—the least accurate encoder (Prithvi, mAP 0.432) is the most FNR-robust (0.130) and the most accurate is not—but the rank correlation over the full roster is only a weak, non-significant positive (Spearman $\rho = +0.32$, $p=0.28$; Table I)—the strong $+0.90$ inversion visible in our first five encoders regresses toward zero when the roster is expanded, so we report it as directional evidence, not a proven inversion (§III). As the protocol’s fourth evidence axis—a conditional and single-encoder finding that is exactly what the report card’s hidden-worst-class-coverage entry is designed to catch—marginal coverage repair can hide a per-class collapse: on EuroSAT, after target-side recalibration restores the *marginal* guarantee, one land-cover class is still covered at only 0.850 under a 0.956 marginal—a failure invisible to accuracy and to ECE, reported as a single load-bearing existence proof, not a population result (Table II). The protocol’s fifth, secondary and descriptive evidence axis is an encoder coverage-debt ranking: the magnitude of geographic coverage debt ranks the frozen GFMs (Prithvi > DOFA (25) > SSL4EO-DINO > SSL4EO-MAE > Clay)—though, because the source-split predictor is in the singleton regime, this ordering coincides with the shifted-accuracy ordering and is reported as descriptive, not as evidence the conformal layer beats accuracy. We establish these on a real EuroSAT-MS (1) cross-region split with five frozen GFMs (the original four plus DOFA (25), added on the EuroSAT grid), corroborate them on GEO-Bench BigEarthNet-S2 (26) (a documented dominant-label proxy, a rigorous per-label multi-label conformal analysis, and a conformal-risk-control arm), and—to test generalization beyond Europe—on the non-European, multi-continent GEO-Bench m-so2sat benchmark. Our analysis is preregistered; where the preregistered *universal*-restoration criterion was not met we preserve that outcome verbatim (§V) and report the conditional, encoder-dependent result as its honest reframe. Fig. 1 summarizes the audit pipeline end to end.

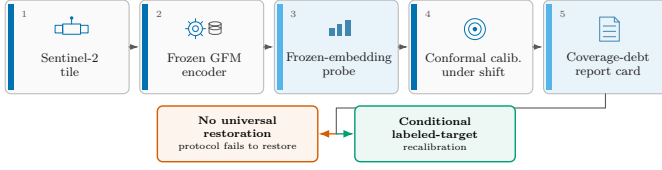


Fig. 1. Overview of the pre-deployment reliability report-card protocol (§IV-B, Table XI). A Sentinel-2 tile (EuroSAT-MS / GEO-Bench BigEarthNet-S2 / m-so2sat) is passed through one of the frozen, interchangeable GFM encoders under audit (five illustrated: Prithvi, Clay, SSL4EO-DINO, SSL4EO-MAE, DOFA) and a frozen-embedding probe (logistic for the single-label tasks, multi-label otherwise), then conformally calibrated under the real geographic (region-transfer) shift—source split-conformal vs. spatial-Mondrian target-region calibration, source = E/Central Europe, target = W/Iberian-Atlantic Europe—yielding per-class conformal sets. The protocol reads off four practitioner-facing numbers for any frozen GFM: the in-distribution FNR guarantee, the geographic debt under shift, the target-slice repair cost, and the hidden worst-class coverage a marginal guarantee alone would miss. The protocol fails to restore coverage universally; the surviving path is conditional, encoder-dependent labeled-target recalibration (§V).

II. METHODS

The data come from EuroSAT-MS (1) (Sentinel-2 L2A, 13 bands, 27,000 tiles, 10 classes). The shift is a *real* geographic cross-region holdout using per-tile longitude: source = E/Central Europe ($\text{lon} \geq P_{33}$), target = W/Iberian-Atlantic Europe ($\text{lon} < P_{33}$); no image transform is applied (Fig. 2; patch-level examples in Fig. 3). The P_{33} cut is a single, fixed longitude-percentile boundary chosen once in advance; a post-hoc boundary-sensitivity sweep over $P_{25}/P_{40}/P_{50}$ is reported in Results (Table V; see Limitations).

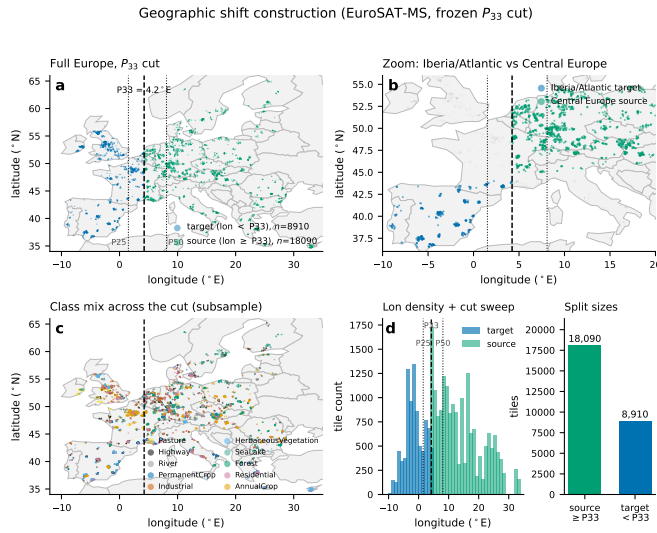


Fig. 2. The geographic-shift construction, shown on the map it is defined on: EuroSAT-MS tile centroids (4,000 of 27,000, seeded subsample) colored by split side at the P_{33} longitude cut ($= 4.2^\circ\text{E}$; P_{25}/P_{50} sweep boundaries dotted). Source tiles (E/Central Europe, $n=18,090$) calibrate; target tiles (W/Iberian-Atlantic Europe, $n=8,910$) are the deployment region. No image transform is applied — the shift is purely geographic.

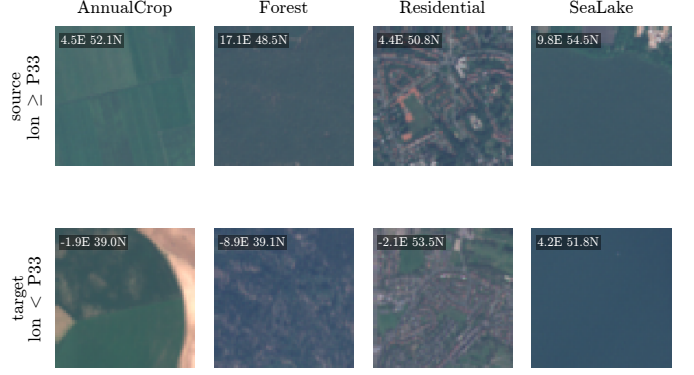


Fig. 3. The shift at patch level: true-color (B04/B03/B02) Sentinel-2 tiles for four land-cover classes, drawn from the source (top row) and target (bottom row) sides of the P_{33} cut with tile centroids annotated. Region-specific crop structure, settlement texture, and water color differ systematically across the cut — the shift is real but subtle at single-patch level, which is exactly why an accuracy-only deployment check can miss it (Results).

This cut also induces a class-frequency (label-prior) shift between source and target that we did not adjust for in the headline logistic probe: at P_{33} , class 5 goes from 2.62% of source tiles to 17.13% of target tiles ($6.54\times$ enriched), class 1 from 15.02% to 3.16% ($\sim 4.7\times$ depleted), class 8 from 11.60% to 4.51%, and class 9 from 14.10% to 5.04% (remaining classes shift by $< 2.4\times$; Table IX). We re-fit the probe with class-balanced sample weighting on the same cached embeddings as a control (Table X; see Discussion). The frozen encoders are Prithvi-EO-2.0-300M (27) (mean-pooled patch tokens, 1024-d); Clay v1.5 (28) (cls token, 1024-d); SSL4EO-S12 (29) ViT-S/16 DINO via torchgeo (cls token, 384-d); SSL4EO-S12 (29) ViT-B/16 MAE (wangyil111/SSL4EO-S12 on HuggingFace, 87.8M params, token-free global pool, 768-d). A fifth encoder, DOFA (Dynamic One-For-All) (25)—a ViT-B/16 whose dynamic wavelength-conditioned patch embedding ingests the Sentinel-2 bands natively (dofa_base_patch16_224, DOFA_MAE weights via torchgeo)—was added on the EuroSAT grid only, frozen and probed identically. The original four weights loaded strictly (zero missing keys), verified real (in-distribution probe accuracy 0.95–0.99 vs. chance 0.10). For the BigEarthNet FNR analysis—the one cross-encoder comparison that turns on statistical power—we expand this battery to *thirteen* frozen encoders by adding eight more loaded through first-party trusted paths (torchgeo 0.9.0 built-in weight builders, no third-party repository code executed), each with normalization read from the weight’s own shipped transform: SSL4EO-S12 MoCo (ViT-S/16, 384-d) and its ResNet50 MoCo/DINO variants (2048-d) (29); SoftCon (30) ViT-B/14 (768-d) and ViT-S/14 (384-d); an SSL4EO MAE ViT-L/16 (1024-d); and CROMA (31) Base (768-d) and Large (1024-d). Two further planned arms (dofa_large, prithvi_600m) were dropped by the pre-specified smoke/mAP gates (missing or non-finite feature caches at extraction), leaving thirteen scored; the three anchor encoders reused the frozen caches and reproduce the earlier five-encoder records exactly. The expanded roster is used only for the BigEarthNet CRC comparison; EuroSAT and m-so2sat retain the five-encoder battery. A logistic probe is trained

on frozen embeddings. We compare source-fit split-conformal (22, 23) (LAC (32) nonconformity $1 - \hat{p}_{\text{true}}$) against a *spatial-Mondrian* conformal predictor (33) that draws per-spatial-bin quantiles from a disjoint same-region target-calibration slice. We adopt the LAC score for its simplicity; adaptive prediction sets (34) are a natural alternative nonconformity score we do not explore. The Mondrian and target-calibration arms rely on exchangeability within each spatial/target bin, a weaker requirement than global exchangeability; more general non-exchangeable-shift theory (35) formalizes how conformal coverage can degrade under shifts broader than the fixed geographic partition studied here, and the conformal-risk-control arm we use below (24) is itself an instance of the general distribution-free risk-controlling framework of Bates et al. (36). We sweep $\alpha \in \{0.05, 0.10, 0.20\}$, 5 seeds, and bootstrap the paired per-seed restoration statistic $|\text{cov}_{\text{split}} - \text{nom}| - |\text{cov}_{\text{Mondrian}} - \text{nom}|$ ($\text{CI} > 0 \Rightarrow \text{restored}$). The set-size cost of that restoration—and the EuroSAT singleton floor it rises above—is reported in Fig. 11. A post-hoc boundary-sensitivity sweep repeats the full protocol at $P_{25}/P_{33}/P_{40}/P_{50}$ longitude cuts on the same cached embeddings, and adds a third, *label-access ablation* arm: non-spatial split conformal whose single LAC quantile is drawn from the same labeled target-region calibration slice the Mondrian arm uses (isolating target-label access from spatial conditioning). All figures (Fig. 9–11, 12, 13, 2, 3, 7, 5, 6, 8, 14) are generated directly from the frozen result records by the accompanying figure-generation scripts. As a post-hoc, exploratory extension, after the preregistered gate (§V) we added three decoupling analyses on the same frozen-feature caches, computed from cached embeddings with no GPU re-extraction: (i) on BigEarthNet-S2, a per-encoder comparison of the CRC arm’s shifted FNR against multi-label accuracy (macro mAP) and calibration (mean per-label ECE) analogs on the same footing (the recomputed shifted FNR reproduces the frozen CRC JSON, e.g. Prithvi 0.129 vs. 0.130); (ii) on EuroSAT, per-class conditional coverage of the spatial-Mondrian-repaired target sets at matched marginal coverage (≈ 0.95 – 0.96), summarized by the worst-class gap $((1-\alpha) - \min_k \text{cov}_k)$; and (iii) a low-shot sweep that subsamples the EuroSAT probe-training fraction to $\{0.01, \dots, 1.0\}$ and records split set size and the coverage–accuracy gap, mapping the singleton-degeneracy boundary. Correlations across encoders are Spearman rank statistics ($n=13$ FNR on the expanded BigEarthNet roster, $n=5$ conditional coverage) and are reported as directional evidence, not proofs of independence (§V).

III. RESULTS

In plain terms: every frozen encoder loses reliability under a real geographic move, that loss is repaired only with labeled target-region data, and the sections below map exactly where a conformal audit adds information beyond accuracy and where it does not. A conformal reliability audit is worth its complexity only if it tells a practitioner something accuracy and calibration do not, so we lead with the regimes in which it does and with the one cross-encoder result that is adequately powered. Primary result (next): under a multi-label false-negative-rate (FNR) risk-control guarantee, all thirteen frozen

encoders incur a real geographic FNR debt that only a labeled target slice repairs, and that debt is bought back with larger prediction sets (the powered FNR-versus-set-size tradeoff); accuracy itself is only a weak, directional guide to which encoder is most robust. We then map the audit’s validity boundary, a deterministic property: in the marginal single-label singleton regime the set-size axis is a restatement of accuracy, so EuroSAT at full data is a mapped negative control, not a decoupling. Next, a remedy-failure result: the textbook unlabeled covariate-shift fix does not close the debt (its weights’ effective sample size collapses), so only a labeled target-region slice reliably restores the guarantee. As a conditional, single-encoder finding: on EuroSAT, restoring the *marginal* coverage guarantee hides a per-class conditional-coverage collapse that neither accuracy nor ECE predicts. Only then do we report the encoder coverage-debt ranking and its target-side recalibration repair, as a secondary, descriptive diagnostic. Every axis-decoupling analysis below is post-hoc and exploratory (added after the preregistered gate; §V).

A. Primary result: a marginal FNR guarantee incurs a geographic debt no encoder escapes, bought back with larger prediction sets—accuracy is only a weak, directional guide to which encoder is most robust

On GEO-Bench `m-bigearthnet` (BigEarthNet-S2; 7,669 tiles, 39 active labels; real E→W geographic shift), a conformal-risk-control (CRC) predictor (24) bounds the example-averaged FNR—a set-valued guarantee with *no top-1-accuracy analog by construction*. We audit *thirteen* frozen encoders here (the five-encoder battery plus eight added at pull expressly to power the cross-encoder comparison; roster in §II). The within-encoder finding is unambiguous and requires no cross-encoder statistics: for *every one* of the thirteen encoders, source-calibrated CRC meets the target FNR in-distribution (0.051–0.056 at $\alpha=0.05$) yet incurs a real FNR debt under shift (0.130 Prithvi to 0.234 ResNet50-MoCo, i.e. 2.6 – $4.7\times$ nominal across the whole roster), which a target-side Mondrian-CRC arm repays (0.002–0.036; Table I, full per- α CRC/repair numbers in Table VI). Unlike the inert EuroSAT-singleton regime (validity-boundary result below), the repair is active here. This debt-and-repair is the reliability-relevant result: the marginal guarantee a practitioner would trust degrades under a regional move—uniformly, across every backbone we tested—and is restored only with a labeled target-region slice. This pattern is not merely descriptive: treating the thirteen preregistered per-encoder point estimates as a population, a distribution-free sign test confirms that shifted CRC FNR exceeds the nominal target in all thirteen encoders (two-sided $p \approx 2.4 \times 10^{-4}$) and that target-side Mondrian-CRC returns FNR to at or below nominal in all thirteen (same test, same p); a Wilcoxon signed-rank sensitivity check agrees exactly (one-sided $p \approx 1.2 \times 10^{-4}$ for both directions, the smallest value attainable at $n=13$), and every one of the thirteen per-encoder bootstrap confidence intervals excludes nominal in the claimed direction, for both the debt and the repair arm. We report the one-sided figure only as a sensitivity

check: the debt-and-repair direction was not discovered in this roster run but inherited from the original five-encoder confirmatory arm, so it is defensible, but the two-sided value is what we use as the headline since it needs no such defense and is already decisive. This is a population-level summary computed post-hoc over the frozen, preregistered per-encoder estimates—the per-encoder numbers are prereg and frozen, but the population test itself is not—and it supplies a second firm, statistically-tested positive result alongside the preregistered falsification of universal restoration (§V). Two further qualifications bound what the test shows. First, the thirteen encoders share the same target-shift BigEarthNet split and labels, so this is a distribution-free summary over thirteen preregistered per-encoder point estimates, not an i.i.d.-sample hypothesis test: the nominal p treats the sign pattern combinatorially, and shared-target dependence means the effective number of independent draws is smaller than thirteen. Second, the repair-arm direction is what the target-calibrated Mondrian-CRC guarantee is constructed to deliver, so its 13/13 reads as a verification that the guarantee holds in practice rather than as an independent discovery. Re-aggregating to the level of independent encoder families—encoders sharing a pretraining corpus and method lineage (all six SSL4EO-S12-pretrained ViT/ResNet variants as one family; SoftCon and CROMA each one family across their two size variants; Prithvi, Clay, and DOFA as singleton families, six families in total) and requiring unanimous within-family agreement, the most conservative vote rule available—still gives a unanimous 6/6 sign test in both directions, but at a far weaker two-sided $p \approx 0.031$ (one-sided $p \approx 0.016$): the family-clustered correction survives at $\alpha=0.05$, but only barely, so the thirteen per-encoder bootstrap confidence intervals (all excluding nominal in the claimed direction) remain the primary evidence, not the population p -value at either n .

The one adequately powered cross-encoder result is a risk-versus-efficiency tradeoff: FNR robustness is bought with larger prediction sets ($\text{Spearman}(\text{FNR}_{\text{sh}}, \text{set size}) = -0.33$ at $\alpha=0.05$, -0.66 ($p=0.017$) at $\alpha=0.10$; Prithvi buys the best FNR with the largest sets, 15.5 labels/tile)—an operating-point choice a practitioner can act on directly, independent of which encoder they picked.

Does accuracy also tell a practitioner *which* encoder to trust? Here the evidence is weaker, and we report it as directional context rather than a claim. The FNR under shift is only weakly rank-correlated with in-distribution mAP ($\text{Spearman}(\text{FNR}_{\text{sh}}, \text{mAP}) = +0.32$, permutation $p=0.28$, $n=13$, $\alpha=0.05$; Fig. 4) and not at all with calibration ($\text{Spearman}(\text{FNR}_{\text{sh}}, \text{ECE}_{\text{in}}) = -0.16$, $p=0.60$). The correlation is *positive*—higher accuracy never buys better FNR robustness, and the ordering never reverses to “better encoder $\rightarrow$ better reliability”—but it does not reach significance and 48 of 78 encoder pairs (62%) invert, so we do *not* claim a strong accuracy-to-reliability inversion; at $n=13$ this is a weak, directional statement, not proof that accuracy is uninformative. One extreme survives the expansion and carries the practical point: Prithvi, the least accurate encoder (mAP 0.432), is the *most* FNR-robust (0.130), while the two most

FNR-fragile encoders—the ResNet50 CNNs (MoCo/DINO, 0.234/0.232)—sit mid-pack on accuracy; and Prithvi vs. Clay is a near-calibration-matched dissociation (ECE 0.082 vs. 0.078, differing by 0.004, yet FNR 0.130 vs. 0.176, the worse-calibrated encoder holding the better FNR). In those first five encoders the same mAP-correlation statistic read $+0.90$ (asymptotic $p=0.037$; its *exact* permutation p was already 0.083); the roster expansion regresses it toward zero, precisely the power test the expansion was run to perform, and we report the weakened result rather than the headline it displaced (§V).

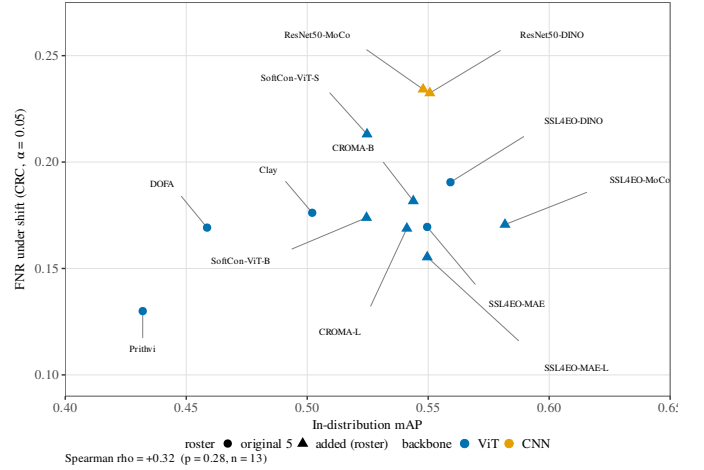


Fig. 4. The five-encoder inversion does not survive expansion to thirteen. FNR under shift vs. in-distribution mAP on m-bigearthnet ($\alpha=0.05$, CRC arm); circles are the original five encoders, triangles the eight added at pull. The rank correlation is a weak, non-significant positive (Spearman = $+0.32$, permutation $p=0.28$, $n=13$; no trend line is drawn, since the correlation does not reach significance); accuracy never buys better FNR robustness, but the strong $+0.90$ inversion of the five-encoder slice regresses toward zero. Prithvi (least accurate) remains the most FNR-robust; the two ResNet50 CNNs are the most fragile at mid accuracy.

TABLE I

PRIMARY DECOUPLING ON GEO-BENCH $M-BIGEARTHNET$ ($\alpha=0.05$; 13 FROZEN ENCODERS, SORTED BY MAP): EVERY ENCODER MEETS THE TARGET FNR IN-DISTRIBUTION AND INCURS A $2.6\text{--}4.7\times$ FNR DEBT UNDER SHIFT THAT THE TARGET-SIDE MONDRIAN-CRC ARM REPAIRS. “FNR SHIFT” IS THE SOURCE-CALIBRATED CRC ARM’S SHIFTED FNR; “REPAIRED” IS THE TARGET-SIDE MONDRIAN-CRC ARM; SET SIZE IS CRC LABELS/TILE UNDER SHIFT. ACCURACY IS A WEAK, NON-SIGNIFICANT GUIDE TO FNR ROBUSTNESS: $\text{Spearman}(\text{FNR}_{\text{sh}}, \text{mAP}) = +0.32$ ($p=0.28$; 48/78 INVERSIONS), $(\text{FNR}_{\text{sh}}, \text{ECE}) = -0.16$ ($p=0.60$), $(\text{FNR}_{\text{sh}}, \text{SET SIZE}) = -0.33$ ($n=13$; -0.66 , $p=0.017$ AT $\alpha=0.10$).

Encoder	mAP (in)	ECE (in)	FNR shift	FNR repaired	Set size
Prithvi	0.432	0.082	0.130	0.031	15.5
DOFA	0.459	0.057	0.169	0.021	13.0
Clay	0.502	0.078	0.176	0.022	13.3
SoftCon-ViT-B	0.525	0.033	0.174	0.031	11.9
SoftCon-ViT-S	0.525	0.020	0.213	0.036	10.7
CROMA-L	0.541	0.037	0.169	0.024	11.6
CROMA-B	0.544	0.046	0.182	0.030	12.1
ResNet50-MoCo	0.548	0.073	0.234	0.002	12.5
SSL4EO-MAE	0.550	0.037	0.170	0.030	11.8
SSL4EO-MAE (ViT-L)	0.550	0.038	0.155	0.029	11.9
ResNet50-DINO	0.551	0.055	0.232	0.014	10.8
SSL4EO-DINO	0.559	0.028	0.191	0.032	10.6
SSL4EO-MoCo	0.582	0.032	0.171	0.028	10.7

B. Secondary (conditional) result: marginal coverage repair hides a per-class conditional collapse

After the target-side spatial-Mondrian arm restores the marginal EuroSAT guarantee to $\approx 0.95\text{--}0.96$ for every encoder (secondary results below), we ask what that marginal scalar hides: how badly does the worst individual land-cover class still under-cover? At matched marginal coverage ($\alpha=0.05$, 5 seeds; Table II; Fig. 5) the worst-class gap is scrambled relative to accuracy and ECE— $\text{Spearman}(\text{worst gap}, \text{acc}) = +0.30$ ($p=0.62$), $\text{Spearman}(\text{worst gap}, \text{ECE}) = -0.30$ ($p=0.62$), 6 of 10 pairwise inversions against each, partial correlation ≈ 0 (-0.03 for $\text{acc}|\text{ece}$). The smoking gun: SSL4EO-DINO is mid-pack on accuracy (0.880, 3rd of 5) and ECE (0.076, 3rd) yet silently abandons a class—one EuroSAT class is covered at only 0.850 under a 0.956 marginal (a 0.106 within-encoder conditional shortfall, $2\times$ the next-worst encoder’s gap), and it is the same class that under-covers in all five seeds (per-seed coverage 0.826–0.871, std 0.016; the datum is seed-stable, not a single-seed artifact). Nothing in its accuracy or calibration predicts this: DOFA, the second-least accurate encoder (0.851), has the best conditional coverage (gap 0.031), while Clay—the most accurate (0.918) and best-calibrated (ECE 0.057)—has the second-worst (gap 0.048). The effect sharpens at $\alpha=0.10$, where SSL4EO-DINO’s worst class sits at 0.776 under a 0.926 marginal. Which frozen encoder quietly drops a class under geographic shift is invisible to accuracy and to ECE and is surfaced only by the conditional conformal audit. Honest caveat: $n=5$ gives low power—all $p > 0.28$ and the weak rank correlation even flips sign across α —so the claim is “no detectable/stable dependence plus large concrete inversions,” not proven independence; the SSL4EO-DINO case is the load-bearing single datum and is reported as such (§V).

TABLE II

PER-CLASS CONDITIONAL COVERAGE OF THE MARGINALLY REPAIRED EUROSAT PREDICTOR (SPATIAL-MONDRIAN, $\alpha=0.05$; MARGINAL COVERAGE MATCHED TO ≈ 0.95 – 0.96 ; 5 SEEDS). THE WORST-CLASS GAP DOES NOT TRACK ACCURACY OR ECE (Spearman = $+0.30 / -0.30$, $p=0.62$; 6/10 INVERSIONS). SSL4EO-DINO, MID-PACK ON BOTH SCALARS, ABANDONS A CLASS AT 0.850 (SEED-STABLE; RANGE 0.826–0.871); THE MOST ACCURATE AND BEST-CALIBRATED ENCODER (CLAY) HAS THE SECOND-WORST GAP. ACCURACY/ECE COLUMNS ARE THE CANONICAL SHIFT VALUES; THE CONDITIONAL COLUMNS ARE THE EXP0 RECOMPUTE.

Encoder	Acc. shift	ECE shift	Marg. cov	Worst-class cov	Worst gap (vs. nominal)
DOFA	0.851	0.093	0.951	0.919	0.031
SSL4EO-MAE	0.898	0.065	0.962	0.912	0.038
Prithvi	0.833	0.109	0.955	0.905	0.045
Clay	0.918	0.057	0.960	0.902	0.048
SSL4EO-DINO	0.880	0.076	0.956	0.850	0.100

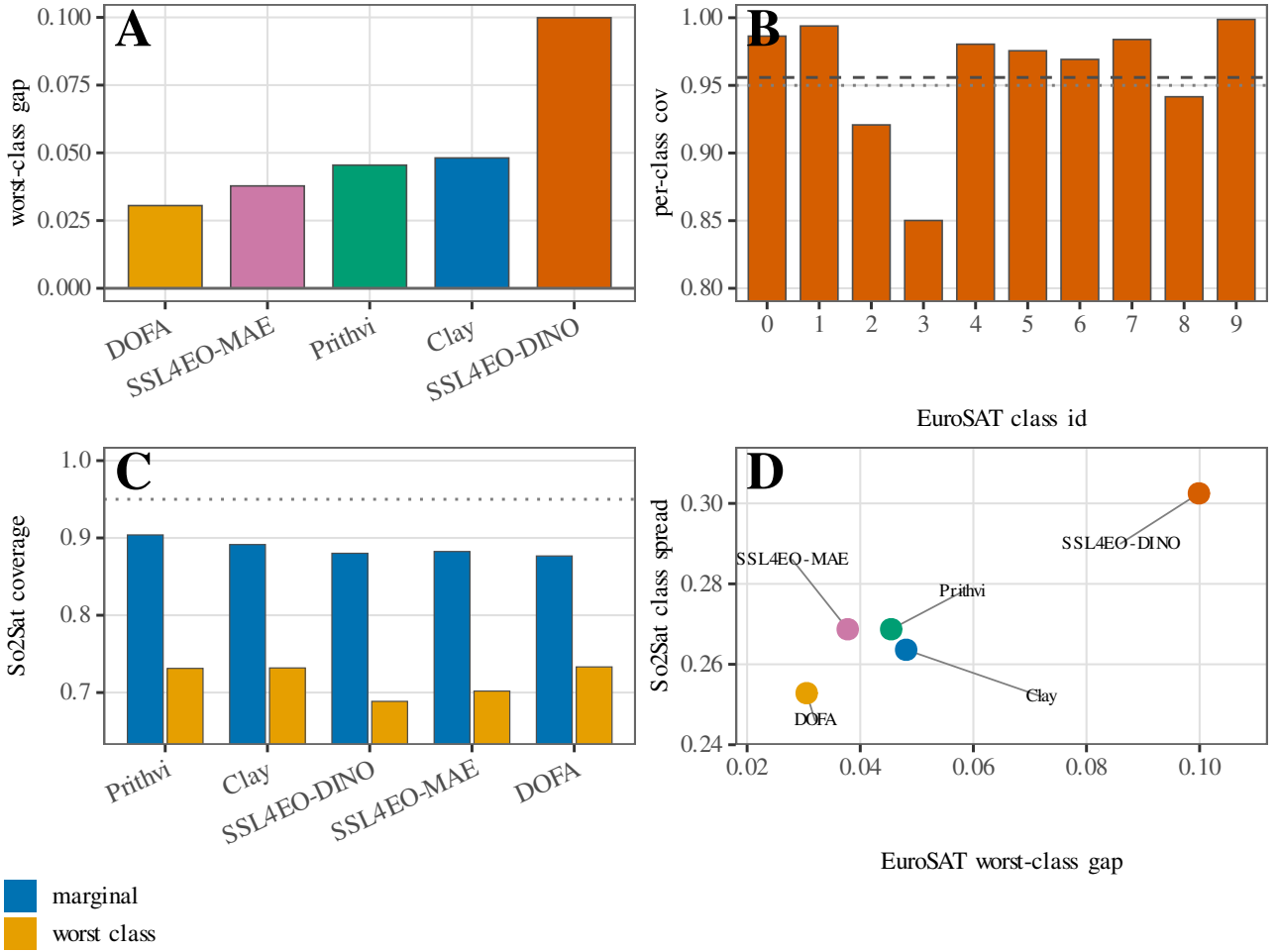


Fig. 5. Conditional-coverage and association panels from the frozen exp0 / So2Sat roster summaries (Tables II and VIII; no new runs). (a) EuroSAT worst-class coverage gap at $\alpha=0.05$ by encoder. (b) Prithvi per-class conditional coverage (dashed = encoder marginal; dotted = nominal 0.95). (c) So2Sat dissociation: shift accuracy vs. worst-class coverage (frozen roster points). (d) EuroSAT worst-gap Spearman / partial associations at $\alpha=0.05$.

C. The audit's validity boundary: singleton degeneracy of the marginal single-label set

A conformal set-size axis can decouple from accuracy only where the sets are non-trivial. On EuroSAT at full data the source-calibrated split predictor emits *singletons* on the shifted target (mean set size 1.00 for every encoder and α ; per-seed ≤ 1.001 ; Fig. 11b), so its coverage coincides with shifted top-1 accuracy (5-seed-mean $|\Delta| \leq 2 \times 10^{-4}$) and the marginal set-size/efficiency axis carries no information beyond accuracy. Even after target-side repair the restored Mondrian set size is a deterministic image of accuracy loss (Fig. 11a): across the five encoders $\text{Spearman}(\text{set size}, \text{acc}) = -1.0$ and $\text{Spearman}(\text{set size}, \text{ECE}) = +1.0$ ($\alpha=0.05$, zero pairwise inversions). We therefore report set size only as a companion to a held guarantee, never as evidence that the conformal layer beats accuracy—the preregistered “efficiency-decoupling” route is empirically falsified on this benchmark and we say so. Mapping the boundary. Subsampling the probe-training fraction drives shifted accuracy down and locates where the audit switches from degenerate to informative (Table III, Fig. 6). In the high-accuracy regime ($\text{acc} \gtrsim 0.80$ at $\alpha=0.05$) sets are exact singletons ($\text{size} \equiv 1.000$) and coverage $\equiv$ accuracy; as the task hardens below $\text{acc} \approx 0.78$ the sets go non-singleton ($\text{size} > 1.05$) and the coverage–accuracy gap opens from 0 to $+0.22$. The more accurate the encoder, the higher the accuracy at which it degenerates (Clay singleton by $\text{acc} \approx 0.87$, SSL4EO-DINO by ≈ 0.82). This makes EuroSAT at full data an explicit negative control: it sits above the boundary, so its *marginal* single-label audit is accuracy-restated—which is precisely why the per-class conditional audit above (on the non-singleton repaired sets) and the multi-label FNR audit on BigEarthNet are where the conformal layer earns its keep, and why So2Sat (17-class LCZ, shifted accuracy 0.52–0.60), which sits below the boundary, carries informative non-singleton sets even marginally (Table VII). The boundary is itself a contribution: it tells a practitioner when a marginal conformal audit of a frozen GFM is worth running.

TABLE III

SINGLETON-DEGENERACY BOUNDARY (EUROSAT, PRITHVI, $\alpha=0.05$, 3 SEEDS): SUBSAMPLING THE PROBE-TRAINING FRACTION LOWERS SHIFTED ACCURACY AND MOVES THE SPLIT PREDICTOR FROM DEGENERATE (SINGLETONS, COVERAGE $\equiv$ ACCURACY) TO INFORMATIVE (NON-SINGLETON SETS, COVERAGE DECOUPLED FROM ACCURACY). THE TRANSITION SITS NEAR ACCURACY ≈ 0.78 – 0.80 .

Train frac	Train n	Shifted acc	Split set size	Coverage – accuracy
1.00	10854	0.832	1.000	+0.000
0.50	5427	0.805	1.026	+0.011
0.25	2713	0.783	1.072	+0.026
0.10	1085	0.765	1.202	+0.066
0.05	542	0.721	1.383	+0.107
0.02	217	0.645	1.866	+0.196
0.01	108	0.650	2.294	+0.221

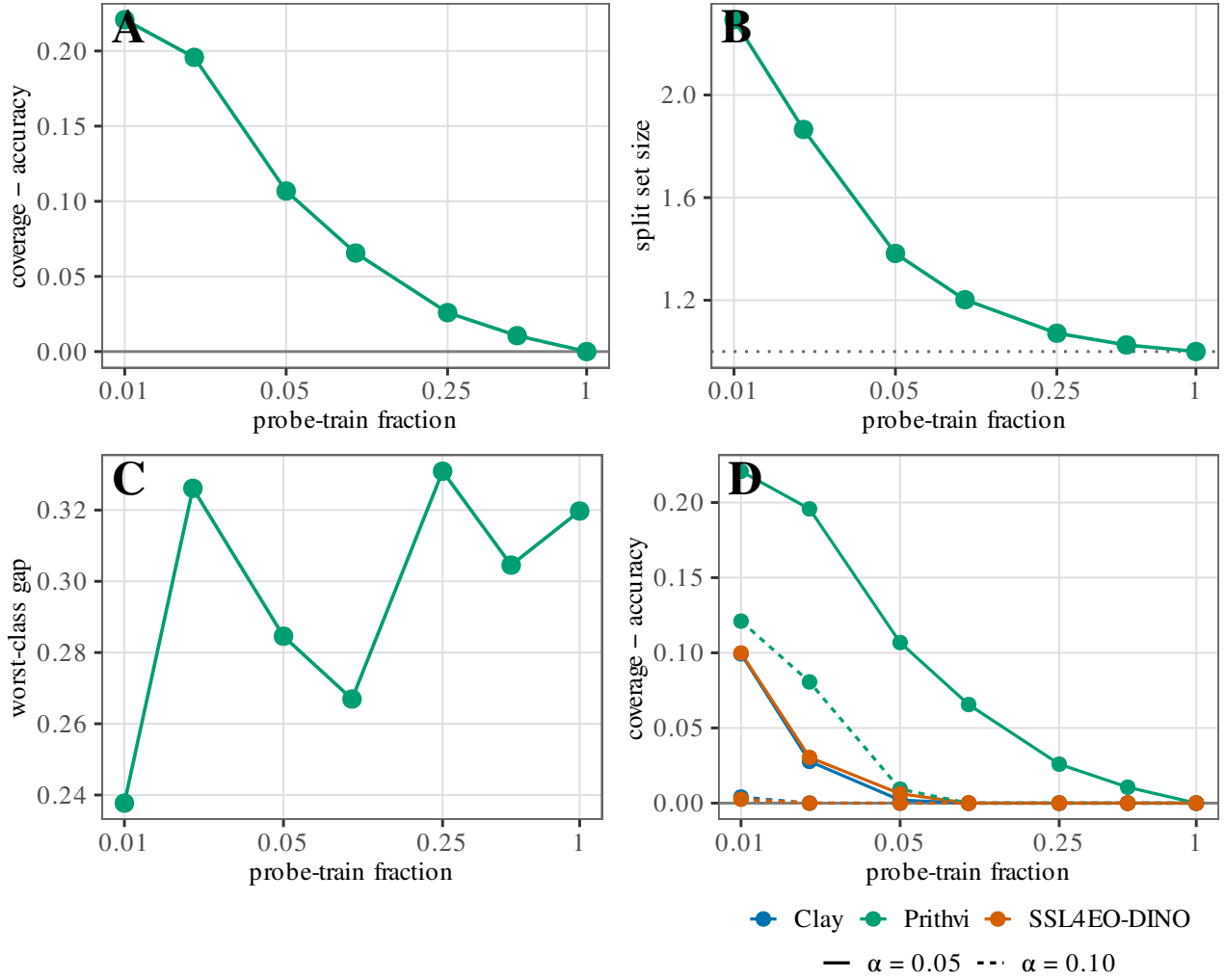


Fig. 6. Four-panel singleton / low-shot boundary from the frozen EuroSAT lowshot record (Table III; no new runs). (a) Coverage–accuracy gap vs. probe-training fraction at $\alpha=0.05$. (b) Mean set size vs. fraction (singleton floor = 1). (c) Worst-class gap vs. fraction. (d) $\alpha=0.05$ vs. 0.10 coverage–accuracy gap at the 10% budget.

D. Secondary descriptive: the coverage-debt encoder ranking and its target-side repair

The remaining EuroSAT results are the coverage-debt ranking and target-recalibration repair. They are a useful descriptive diagnostic but, per the validity-boundary result above, the marginal ranking coincides with the shifted-accuracy ranking; we report them as such, not as an independent conformal signal. Coverage debt is real and encoder-dependent: under shift, accuracy drops $0.95 \rightarrow 0.83$ (Prithvi), $0.97 \rightarrow 0.88$ (SSL4EO-DINO), $0.98 \rightarrow 0.92$ (Clay), and ECE rises to 0.109 (Prithvi), 0.076 (SSL4EO-DINO), 0.065 (SSL4EO-MAE), 0.057 (Clay)—an ordering of calibration degradation across the four encoders that is preserved under the boundary sweep (Table V; Fig. 9C) and the class-prior-balanced probe control (Discussion), so it does not rest on a bare four-point monotonicity (Fig. 9A–B). As a continuous, bucket-free debt measure we integrate the target coverage gap $\max(0, (1 - \alpha) - \text{cov}_{\text{split}})$ over the α sweep: $D = 8.0/3.3/1.4/0.8 \times 10^{-3}$ for Prithvi/SSL4EO-DINO/SSL4EO-MAE/Clay—and $D \approx 6.2 \times 10^{-3}$ for DOFA, added on the same grid as a fifth encoder and the second-heaviest debt, between Prithvi and SSL4EO-DINO—recovering the same ordering without the coarse $k/3$ repaired-levels bucketing. The gap closes at $\alpha = 0.20$ for all five encoders because shifted accuracy already meets the 0.80 nominal—a no-debt artifact of the easy risk level, not evidence of a repair. DOFA’s shifted split coverage is 0.851 at every risk level, so like the four battery encoders it carries under-coverage debt only at $\alpha = 0.05$ and 0.10.

Spatial-Mondrian repays the debt where it exists: at $\alpha = 0.05$ (nominal 0.95), source split-conformal under-covers (0.833 Prithvi, 0.851 DOFA, 0.880 SSL4EO-DINO, 0.898 SSL4EO-MAE,¹ 0.918 Clay) while spatial-Mondrian restores coverage to 0.955, 0.951, 0.956, 0.962, and 0.960 respectively, with paired-bootstrap CI excluding 0 for all five encoders (Fig. 10). The magnitude of repair tracks the debt: at this shared risk level the restoration set size is monotone in debt—1.62 (Prithvi) > 1.58 (DOFA) > 1.43 (SSL4EO-DINO) > 1.39 (SSL4EO-MAE) > 1.22 (Clay) (Fig. 11). The number of risk levels repaired is weakly monotone: Prithvi (heaviest debt)

and DOFA (second-heaviest) are each repaired at 2/3 levels ($\alpha = 0.05, 0.10$), while SSL4EO-DINO, SSL4EO-MAE, and Clay are each repaired at 1/3 ($\alpha = 0.05$). At $\alpha = 0.20$ (nominal 0.80) none of the five is repaired because shifted accuracy (0.83–0.92, DOFA 0.85) already meets nominal—there is no under-coverage debt to repay, and the correction correctly does nothing (Fig. 10). DOFA was added to the same EuroSAT grid as a fifth encoder in a single-FM run whose spatial-Mondrian set sizes are 1.58/1.29/1.08 at $\alpha = 0.05/0.10/0.20$; an aggregate fail under the multi-encoder universal-restoration rule is expected here (Clay and Prithvi were absent from that single-encoder call), while the per-encoder outcome is restoration at $\alpha = 0.05$ and 0.10 by the same paired-bootstrap CI-excludes-0 test applied to the four battery encoders (paired gap-difference CI low > 0), and no repair at $\alpha = 0.20$, where DOFA already over-covers (0.851). Correction note: this DOFA arm was re-run after an initial run inadvertently probed DOFA’s 45-way classification-head output instead of its 768-d pooled feature embedding; all DOFA numbers reported here are from the corrected embedding run, with the incorrect probe outputs archived for provenance. The four battery encoders are unaffected: their cached features were verified at their documented embedding widths (Prithvi and Clay 1024-d, SSL4EO-DINO 384-d, SSL4EO-MAE 768-d), and the bug’s 45-dimensional signature appears in no other feature cache.

Temperature scaling (37) proves insufficient: applying the source-fit temperature to the shifted logits lowers ECE to 0.058/0.041/0.054/0.032 (Prithvi/SSL4EO-DINO/SSL4EO-MAE/Clay) from the raw 0.109/0.076/0.065/0.057, but the residual still sits 2–11 $\times$ above the in-distribution level (0.026/0.014/0.005/0.013), motivating the distribution-free conformal route; a focal-loss-based calibration alternative (38) is likewise post-hoc-parametric and is not explored here.

The headline is not a single-cut artifact: sweeping the source/target longitude cut over $P_{25}/P_{33}/P_{40}/P_{50}$ on the same cached embeddings (5 seeds each; Table V) preserves the shift-ECE encoder ordering Prithvi > SSL4EO-DINO > SSL4EO-MAE > Clay at every boundary (Prithvi 0.144/0.108/0.103/0.093 vs. Clay 0.074/0.057/0.051/0.041 at $P_{25}/P_{33}/P_{40}/P_{50}$), and the debt shrinks monotonically as the cut moves east and the target becomes less extreme (Prithvi split coverage 0.792 $\rightarrow$ 0.859). The ordering is over a small roster (five encoders at P_{33} , four across the sweep) and some adjacent gaps sit within seed noise—e.g. SSL4EO-DINO and SSL4EO-MAE differ by only ≈ 0.001 shift-ECE at P_{25} —so the extremes (Prithvi heaviest, Clay lightest) are robust while adjacent mid-rank ties should not be over-interpreted. At $\alpha = 0.05$, Mondrian restoration (paired-seed CI excluding 0) holds in 15 of 16 encoder $\times$ boundary cells; the sole exception is Clay at P_{50} , where split coverage is already 0.936 (debt ≈ 0.014)—exactly the debt-targeted behaviour the conditional framing predicts. At $\alpha = 0.10$, Prithvi (heaviest debt) is restored at all four boundaries while the lighter-debt encoders mostly are not.

A label-access ablation shows that target calibration, not spatial binning, is the active ingredient: the non-spatial target-calibrated arm restores coverage to 0.946–0.952 at $\alpha = 0.05$ in every debt cell—statistically indistinguish-

¹SSL4EO-MAE’s shifted split coverage is 0.8984 in this five-encoder run (shown as 0.898) and 0.8985 in the covariate- and label-shift baseline runs (Tables IV–IV, shown as 0.899); the $<10^{-4}$ difference is independent calibration-split RNG on the same cached features, and split coverage equals shifted top-1 accuracy in every case (singleton sets).

able from spatial-Mondrian: the paired-seed “Mondrian minus target-global” gap-reduction CI excludes 0 in zero of 48 ($\text{FM} \times \text{boundary} \times \alpha$) cells, while target-global itself beats source-only split-conformal in the 22 cells where debt exists. Mondrian’s set sizes are consistently larger (P_{33} , $\alpha = 0.05$: 1.62 vs. 1.56 Prithvi; 1.44 vs. 1.30 SSL4EO-DINO; 1.39 vs. 1.20 SSL4EO-MAE; 1.22 vs. 1.12 Clay). On this benchmark and 4×4 grid, then, the repair is bought by access to labeled target-region calibration data; per-spatial-bin conditioning adds no measurable marginal coverage benefit. We report this honestly and adjust the claim: the operative correction is *target-side conformal recalibration*, of which spatial-Mondrian is one (slightly costlier) instantiation here; finer spatial heterogeneity, where within-target variation is stronger, may still separate the two arms. Once spatial conditioning is inert, the remedy reduces to standard split-conformal calibrated on a small labeled target-region slice: we introduce no new conformal method, and spatial binning is reported here as an honest negative. The contributions of this paper are correspondingly the encoder-robustness diagnostic and the preregistered audit, not a new estimator.

How small a labeled target-region slice does the repair require under this Cross-European optical shift? The frozen protocol prices the remedy at a single point—the fixed 30% target-calibration fraction—so we sweep the budget post-hoc over 0.5%–30% of the target region on the same cached embeddings, holding the evaluation slice fixed at the frozen cut (Fig. 7, *descriptive, not confirmatory*). The repaired arm clears the nominal level at the smallest swept budget and then *decreases* toward nominal as the budget grows: finite-sample conformal conservatism, not a gradual repair. At $n=44$ the effective quantile $\lceil (n+1)(1-\alpha) \rceil / n$ is 0.932 vs. nominal 0.900 at $\alpha=0.10$, over-covering and paying for the slack in larger sets; as the budget grows toward 30% the inflation shrinks and coverage converges onto nominal—from far above it at the cheapest budget (Prithvi at 0.5%: 0.994/0.959/0.904 at $\alpha=0.05/0.10/0.20$) down to the frozen operating point’s 0.952/0.899/0.833. For a practitioner the labeled budget therefore controls set-size tightness, not coverage validity: wherever debt exists, any labeled slice—down to the finite-sample floor—already repairs the guarantee. The exchange rate is steep: for Prithvi, moving from the frozen $n=2673$ down to $n=44$ —a $61 \times$ annotation saving—inflates the mean set from 1.56 to 3.66 at $\alpha=0.05$ (+134%), from 1.22 to 1.77 at $\alpha=0.10$ (+46%), and from 1.00 to 1.28 at $\alpha=0.20$ (+28%). Clay at $\alpha=0.20$ is flat at the split value 0.9180 at every budget (singleton regime; Table III). The practical consequence is a deflationary one for the 30% figure the report card quotes: it is a *sufficient* budget, not a necessary one. The BigEarthNet CRC arm shows the same structure with a hard floor: CRC can only certify risk α when $\alpha \geq 1/(n_{\text{eff}}+1)$, so at $\alpha=0.05$ at least 19 positive-label calibration examples are needed; below that the set becomes every label and measured FNR collapses to zero vacuously (excluded rather than reported as a free repair). Above the floor, FNR rises toward the target as the budget grows (e.g. at $\alpha=0.05$, 0.022 at 1% against 0.051 at 30%), while $\approx 1\%$ of the target region already discharges the risk debt and the remaining annotation buys tighter sets.

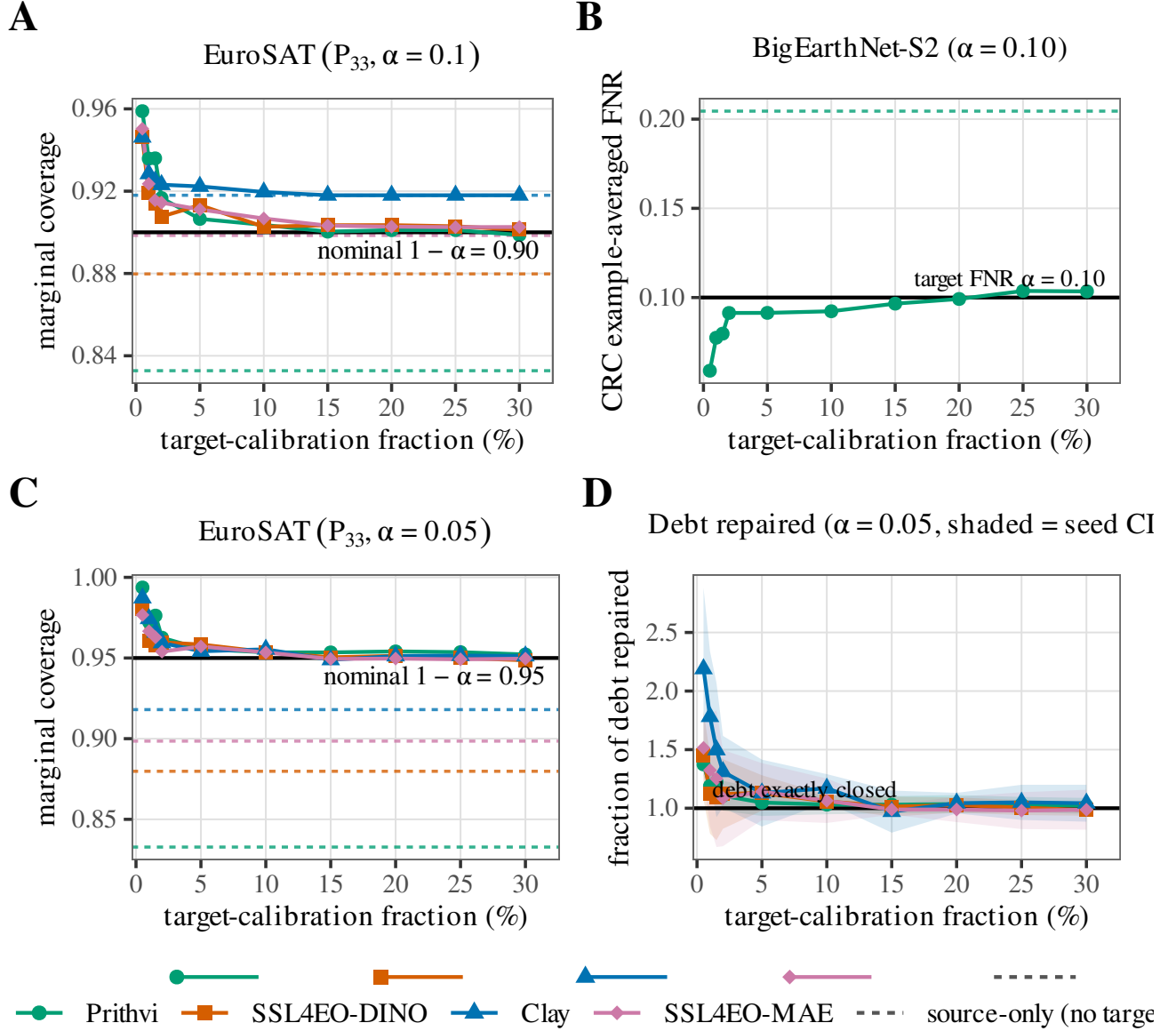


Fig. 7. Target-recalibration planning curves: annotation budget versus guarantee tightness. (A) EuroSAT-MS (P_{33} , $\alpha=0.10$): marginal coverage of the target-calibrated arm versus the labeled fraction of the target region (%). (B) GEO-Bench BigEarthNet-S2 ($\alpha=0.10$): CRC example-averaged false-negative rate (FNR) on the same x -axis (Prithvi only; vacuous CRC budgets with $\alpha < 1/(n_{\text{eff}}+1)$ omitted). (C) EuroSAT-MS at the stricter $\alpha=0.05$, same coverage axes as (A). (D) Normalized EuroSAT reading at $\alpha=0.05$: fraction of coverage debt repaired versus budget (shaded bands = five-seed intervals; values >1 are over-repair). Solid coloured curves are encoders; matching dashed horizontals are budget-independent source-only baselines (no target labels); black horizontals mark the nominal coverage $1-\alpha$, target FNR α , or exact debt clearance ($=1$ in (D)). The repair is not a rising ramp: wherever debt exists, the smallest usable budget already clears the nominal level, after which curves converge onto it from the conservative side. That shape is finite-sample conformal correction—the effective quantile $[(n+1)(1-\alpha)]/n$ inflates above $1-\alpha$ at small n (e.g. 0.932 vs. 0.900 at $n=44$, $\alpha=0.10$), over-covering and paying for the slack in larger sets—so extra labels buy tightness to nominal, not validity of the guarantee. Descriptive secondary analysis (§V); So2Sat excluded (no local features).

Tables IV–VII are a robustness battery, not five independent findings: each holds the frozen encoders and the coverage debt fixed and swaps exactly one variable—the correction method (unlabeled covariate-shift reweighting in Table IV; label-shift-adjusted reweighting in Table IV), the source/target geographic cut (Table V), or the benchmark itself (BigEarthNet’s multi-label FNR arm in Table VI; the non-European So2Sat benchmark in Table VII). In every table the single comparison worth making is the same one: at the shared $\alpha = 0.05$ row, read the source-only (“split”) figure against the labeled target-side

repair figure in that row—the gap between them is the debt each table is stress-testing, and it is the only number that would change the paper’s claim if it closed.

A stronger unlabeled baseline still does not close the debt. Is the labeled target slice truly necessary, or would a principled unlabeled remedy suffice? As a post-hoc exploratory baseline on the cached EuroSAT features we add covariate-shift-weighted split conformal (17): density-ratio weights $w(x)$ from a logistic source-vs-target discriminator on the frozen embeddings, weighted-quantile calibration on the

source scores, scored on the same shifted target (Table IV). Reweighting recovers part of the debt—at $\alpha = 0.05$ target coverage rises from source-split 0.833–0.918 to 0.911–0.938 per encoder—but under-covers the nominal 0.95 for every encoder and never matches the labeled target-global arm (≈ 0.95). The mechanism is visible in the weights: the effective sample size is only 0.1–1.6% of the calibration set — density ratios so heavy-tailed that a handful of points dominate, a direct signature that this real $E \rightarrow W$ shift is not the pure covariate shift under which weighted conformal is guaranteed (it also carries the class-prior shift documented in Methods). Even a principled unlabeled correction is thus insufficient and unstable here; a small labeled target slice remains the reliable remedy.

A label-shift-adjusted control rules out the class-prior confound: the P_{33} cut’s severe class-prior shift (Table IX) raises the possibility that the coverage debt above is a textbook label-shift effect (39) rather than a genuinely geographic one. We test this directly at the identical P_{33} split: we reweight the source calibration scores by $w(y) = \hat{\pi}_{\text{target}}(y)/\hat{\pi}_{\text{source}}(y)$ (Podkopaev–Ramdas label-shift-weighted split conformal), in an *oracle* variant (the true target-region label marginal, an upper bound on what prior-shift correction alone could repay) and a deployable variant (target priors from confusion-matrix Black-Box Shift Estimation (40), using unlabeled target predictions only). Neither restores coverage. For three of four encoders both variants make coverage worse than doing nothing (Table IV), and for Prithvi (heaviest debt) both recover only part of the gap while remaining far below nominal and the labeled target-global arm. The overcorrection compounds at looser risk levels—at $\alpha = 0.20$ every encoder’s label-shift-adjusted coverage sits 8.1–17.5 pp below nominal—and more than 15 pp below even the do-nothing source-split arm.² Because label-shift weighting is provably valid only when $P(x | y)$ is shift-invariant, this systematic failure rules out a pure class-prior-shift explanation and is consistent with a within-class conditional shift in the frozen encoders’ features across regions—the geographic mechanism this paper posits, which target-side recalibration (which assumes nothing about $P(x | y)$) correctly repairs and label-shift reweighting cannot.

E. Replication on a second benchmark (BigEarthNet)

We replicate the phenomenon on a second real EO benchmark: GEO-Bench (26) m-bigearthnet (BigEarthNet-S2 (41); 7,669 usable tiles, real lat/lon, 36.96–67.80°N). We apply two complementary analyses.

A documented dominant-label single-label proxy (12 classes, ≥ 60 tiles each) lets the single-label LAC harness apply unchanged. All three frozen FMs replicate the pattern: ECE degrades (0.36 $\rightarrow$ 0.56 Prithvi, 0.15 $\rightarrow$ 0.42 SSL4EO, 0.33 $\rightarrow$ 0.59 Clay) and spatial-Mondrian restores coverage to near-nominal at all three risk levels (Fig. 12). The proxy is a hard task (frozen-probe accuracy 0.50–0.61 in-distribution,

dropping to 0.27–0.37 under shift), so restoration is bought at large prediction-set sizes (~ 6 –10 of 12 classes)—a coarse, qualitative replication.

We extend the analysis with rigorous per-class binary conformal prediction on the 43-class multi-hot labels. For each of 39 well-populated labels (≥ 120 global positives; 27 active per seed after per-split filtering), we fit an independent binary probe (StandardScaler, L2-logistic, $C = 1, 5$ seeds) on source-region embeddings and calibrate per-label recall-coverage $\Pr(c \in \mathcal{C}(x) | y_c = 1)$ via source-only split-conformal and target-side spatial-Mondrian arms. The coverage-debt phenomenon replicates per-label: mean recall-coverage drops from 0.92 in-distribution to 0.75 under shift ($\alpha = 0.10$; mean per-label debt 0.173), with spatial-Mondrian restoring to 0.91 (per-label paired CI excludes 0 at $\alpha \in \{0.10, 0.20\}$ for all three FMs; Fig. 13). Predicted-label set sizes remain large (20.9 of 27 active labels per tile at $\alpha = 0.10$), reflecting the geographic distributional mismatch and the recall requirement. Per-label conformal is more principled than the dominant-label proxy—it offers independent per-class recall guarantees—but both yield comparably large prediction sets under this hard pan-European shift.

This conformal-risk-control (CRC) arm, which gives a finite-sample FNR guarantee, is the substrate for the paper’s primary decoupling result (Table I); here we give its full per- α calibration and repair numbers for all five frozen encoders. Per-label split-conformal offers per-class recall calibration but no finite-sample guarantee on the example-averaged false-negative rate (FNR). We therefore add a conformal-risk-control (CRC) arm (24) on the same frozen-feature caches, splits, 4×4 Mondrian grid, and per-label probes (per-example FNR loss, monotone in the threshold λ ; exact weighted-quantile calibration; 5 seeds; percentile-bootstrap CIs, $B=2000$). The three-step story repeats under the marginal-FNR guarantee for every encoder (Table VI): (i) source-calibrated CRC meets the target FNR in-distribution (e.g. 0.105 at $\alpha=0.10$, Prithvi); (ii) the same $\hat{\lambda}$ incurs FNR debt under the geographic shift (2–4 $\times$ the nominal level, e.g. 0.206–0.277 at $\alpha=0.10$); (iii) a Mondrian-CRC arm calibrated per spatial bin on the target-region slice restores marginal FNR to at-or-below nominal, at the cost of larger label sets (~ 17 –18 vs. ~ 8 –12 labels/tile at $\alpha=0.10$). An honest efficiency note: CRC controls only the example-averaged FNR, so its in-distribution sets are smaller than per-label split-conformal’s, and in-distribution FNR at $\alpha=0.05$ fluctuates slightly above nominal (mean 0.052–0.056), consistent with the expectation-level $\mathbb{E}[L] \leq \alpha$ guarantee—which holds in expectation over calibration draws, not per individual run, so a single run landing marginally above α is expected, not a violation. Like spatial-Mondrian conformal, Mondrian-CRC requires labeled target-region calibration data—a shift-aware remedy, not a free lunch. The CRC / method comparison is summarized in Fig. 8.

²The label-shift run reproduces the Split and Target-global columns of Table IV within ± 0.001 ; its own covariate-shift-weighted reference column (not shown) differs slightly because the two runs use different, both legitimate, base-rate corrections for the density-ratio discriminator. This affects only that reference column, not the label-shift arms.

TABLE IV

UNLABELED-REMEDY CONTROLS ON THE IDENTICAL EUROSAT E→W SPLIT (5-SEED MEANS; FULL $\alpha \in \{0.05, 0.10, 0.20\}$ SWEEP FROM THE FROZEN RESULT RECORDS): COVARIATE-SHIFT-WEIGHTED SPLIT CONFORMAL (17) (DENSITY-RATIO WEIGHTS FROM A SOURCE-VS-TARGET DISCRIMINATOR ON THE FROZEN EMBEDDINGS; “ESS” IS THE WEIGHTS’ EFFECTIVE SAMPLE SIZE AS A FRACTION OF THE CALIBRATION SET, FIXED BEFORE CALIBRATION AND HENCE CONSTANT ACROSS α) AND LABEL-SHIFT-WEIGHTED SPLIT CONFORMAL (39) (ORACLE: TRUE TARGET LABEL MARGINAL; BBSE (40): TARGET PRIORS FROM UNLABELED TARGET PREDICTIONS). SPLIT, WEIGHTED, ESS, AND TARGET-GLOBAL COLUMNS ARE FROM THE WEIGHTED-RUN RECORD; THE LABEL-SHIFT RECORD REPRODUCES SPLIT AND TARGET-GLOBAL WITHIN ± 0.001 . WEIGHTED CONFORMAL UNDER-COVERS NOMINAL AT EVERY α FOR EVERY ENCODER (THE ESS COLLAPSE SIGNALS THE SHIFT IS NOT PURE COVARIATE SHIFT); LABEL-SHIFT ADJUSTMENT MAKES THREE OF FOUR ENCODERS *worse* THAN DOING NOTHING, AND ITS DEFICIT WIDENS AT LOOSER α . ONLY THE LABELED TARGET-GLOBAL ARM REACHES NOMINAL.

α	Encoder	Split	Weighted	ESS frac.	LS-oracle	LS-BBSE	Target-global
0.05	Prithvi	0.833	0.919	1.6%	0.879	0.884	0.952
0.05	SSL4EO-DINO	0.880	0.938	1.3%	0.858	0.867	0.949
0.05	SSL4EO-MAE	0.899	0.911	0.4%	0.856	0.865	0.950
0.05	Clay	0.918	0.927	0.1%	0.891	0.895	0.951
0.10	Prithvi	0.833	0.881	1.6%	0.793	0.798	0.899
0.10	SSL4EO-DINO	0.880	0.914	1.3%	0.766	0.774	0.901
0.10	SSL4EO-MAE	0.899	0.905	0.4%	0.793	0.803	0.903
0.10	Clay	0.918	0.923	0.1%	0.825	0.832	0.918
0.20	Prithvi	0.833	0.855	1.6%	0.675	0.682	0.833
0.20	SSL4EO-DINO	0.880	0.898	1.3%	0.625	0.639	0.880
0.20	SSL4EO-MAE	0.899	0.902	0.4%	0.688	0.704	0.898
0.20	Clay	0.918	0.921	0.1%	0.711	0.719	0.918

TABLE V

BOUNDARY-SENSITIVITY SWEEP, REAL-SHIFT TARGET COVERAGE “SPLIT / TARGET-GLOBAL / SPATIAL-MONDRIAN” (5-SEED MEANS; SPLIT = SOURCE-ONLY CALIBRATION; TARGET-GLOBAL = NON-SPATIAL, TARGET-CALIBRATED ABLATION ARM; FULL $\alpha \in \{0.05, 0.10, 0.20\}$ SWEEP FROM THE SAME FROZEN RESULT RECORD, EXPANDING THE $\alpha=0.05$ HEADLINE ROW). [†] = MONDRIAN-VS-SPLIT PAIRED CI DOES NOT EXCLUDE 0 (NO SIGNIFICANT RESTORATION). AT $\alpha=0.05$, RESTORATION HOLDS IN 15 OF 16 ENCODER×BOUNDARY CELLS (THE SOLE EXCEPTION, CLAY AT P_{50} , WHERE DEBT IS ≈ 0.014 , IS MARKED); AT LOOSER α MOST CELLS NO LONGER NEED REPAIR BECAUSE SHIFTED ACCURACY ALREADY APPROACHES NOMINAL.

α	Encoder	P_{25}	P_{33} (headline)	P_{40}	P_{50}
0.05	Prithvi	0.792 / 0.946 / 0.952	0.833 / 0.952 / 0.955	0.843 / 0.950 / 0.955	0.859 / 0.949 / 0.952
0.05	SSL4EO-DINO	0.847 / 0.947 / 0.951	0.880 / 0.949 / 0.956	0.892 / 0.950 / 0.955	0.896 / 0.951 / 0.955
0.05	SSL4EO-MAE	0.865 / 0.948 / 0.957	0.898 / 0.950 / 0.962	0.910 / 0.946 / 0.962	0.917 / 0.952 / 0.967
0.05	Clay	0.897 / 0.950 / 0.955	0.918 / 0.951 / 0.960	0.926 / 0.947 / 0.960	0.936 / 0.950 / 0.963 [†]
0.10	Prithvi	0.789 / 0.899 / 0.898	0.833 / 0.899 / 0.911	0.841 / 0.902 / 0.915	0.854 / 0.902 / 0.917
0.10	SSL4EO-DINO	0.847 / 0.897 / 0.906	0.880 / 0.901 / 0.926 [†]	0.892 / 0.901 / 0.928 [†]	0.896 / 0.902 / 0.930 [†]
0.10	SSL4EO-MAE	0.865 / 0.895 / 0.920 [†]	0.898 / 0.903 / 0.934 [†]	0.910 / 0.910 / 0.938 [†]	0.917 / 0.917 / 0.947 [†]
0.10	Clay	0.897 / 0.900 / 0.920 [†]	0.918 / 0.918 / 0.936 [†]	0.926 / 0.926 / 0.935 [†]	0.936 / 0.936 / 0.943 [†]
0.20	Prithvi	0.789 / 0.802 / 0.832 [†]	0.833 / 0.833 / 0.856 [†]	0.841 / 0.841 / 0.867 [†]	0.854 / 0.854 / 0.877 [†]
0.20	SSL4EO-DINO	0.847 / 0.847 / 0.850 [†]	0.880 / 0.880 / 0.888 [†]	0.892 / 0.892 / 0.896 [†]	0.896 / 0.896 / 0.902 [†]
0.20	SSL4EO-MAE	0.865 / 0.865 / 0.880 [†]	0.898 / 0.898 / 0.905 [†]	0.910 / 0.910 / 0.917 [†]	0.917 / 0.917 / 0.929 [†]
0.20	Clay	0.897 / 0.897 / 0.897 [†]	0.918 / 0.918 / 0.918 [†]	0.926 / 0.926 / 0.926 [†]	0.936 / 0.936 / 0.936 [†]

TABLE VI
CRC ARM ON GEO-BENCH M -BIGEARTHNET (MARGINAL PER-EXAMPLE FNR; MEAN [95% BOOTSTRAP CI] POOLED OVER 5 SEEDS). SOURCE-CALIBRATED CRC MEETS NOMINAL IN-DISTRIBUTION, INCURS FNR DEBT UNDER THE REAL $E \rightarrow W$ SHIFT, AND TARGET-SIDE MONDRIAN-CRC RESTORES IT. SET SIZES ARE LABELS/TILE (OF ~ 27 ACTIVE) UNDER SHIFT.

α	Encoder	CRC FNR in-dist	CRC FNR shift	Mondrian-CRC FNR shift	Sets CRC/Mond.
0.05	Prithvi	0.055 [0.051,0.058]	0.130 [0.125,0.135]	0.031 [0.029,0.034]	15.5 / 22.3
0.05	Clay	0.053 [0.049,0.056]	0.176 [0.171,0.182]	0.022 [0.020,0.024]	13.3 / 23.8
0.05	SSL4EO-DINO	0.053 [0.050,0.057]	0.191 [0.185,0.196]	0.032 [0.029,0.034]	10.6 / 22.3
0.05	SSL4EO-MAE	0.052 [0.048,0.055]	0.170 [0.164,0.175]	0.030 [0.028,0.033]	11.8 / 22.1
0.05	DOFA	0.056 [0.052,0.059]	0.169 [0.164,0.175]	0.021 [0.019,0.024]	13.0 / 24.1
0.10	Prithvi	0.105 [0.101,0.110]	0.206 [0.200,0.212]	0.083 [0.079,0.086]	11.9 / 18.2
0.10	Clay	0.108 [0.103,0.113]	0.277 [0.271,0.283]	0.082 [0.078,0.086]	9.7 / 18.2
0.10	SSL4EO-DINO	0.107 [0.102,0.112]	0.277 [0.271,0.283]	0.082 [0.078,0.086]	7.9 / 17.4
0.10	SSL4EO-MAE	0.108 [0.103,0.113]	0.262 [0.256,0.268]	0.084 [0.080,0.088]	8.6 / 17.1
0.10	DOFA	0.107 [0.103,0.112]	0.249 [0.243,0.256]	0.080 [0.076,0.084]	9.9 / 18.1
0.20	Prithvi	0.205 [0.199,0.211]	0.346 [0.339,0.353]	0.184 [0.179,0.190]	8.0 / 12.9
0.20	Clay	0.203 [0.197,0.209]	0.426 [0.419,0.433]	0.185 [0.179,0.190]	6.4 / 12.9
0.20	SSL4EO-DINO	0.206 [0.200,0.212]	0.401 [0.394,0.408]	0.191 [0.185,0.197]	5.3 / 11.1
0.20	SSL4EO-MAE	0.209 [0.202,0.215]	0.408 [0.401,0.415]	0.189 [0.184,0.195]	5.5 / 11.3
0.20	DOFA	0.203 [0.196,0.209]	0.384 [0.377,0.391]	0.187 [0.181,0.193]	6.7 / 12.4

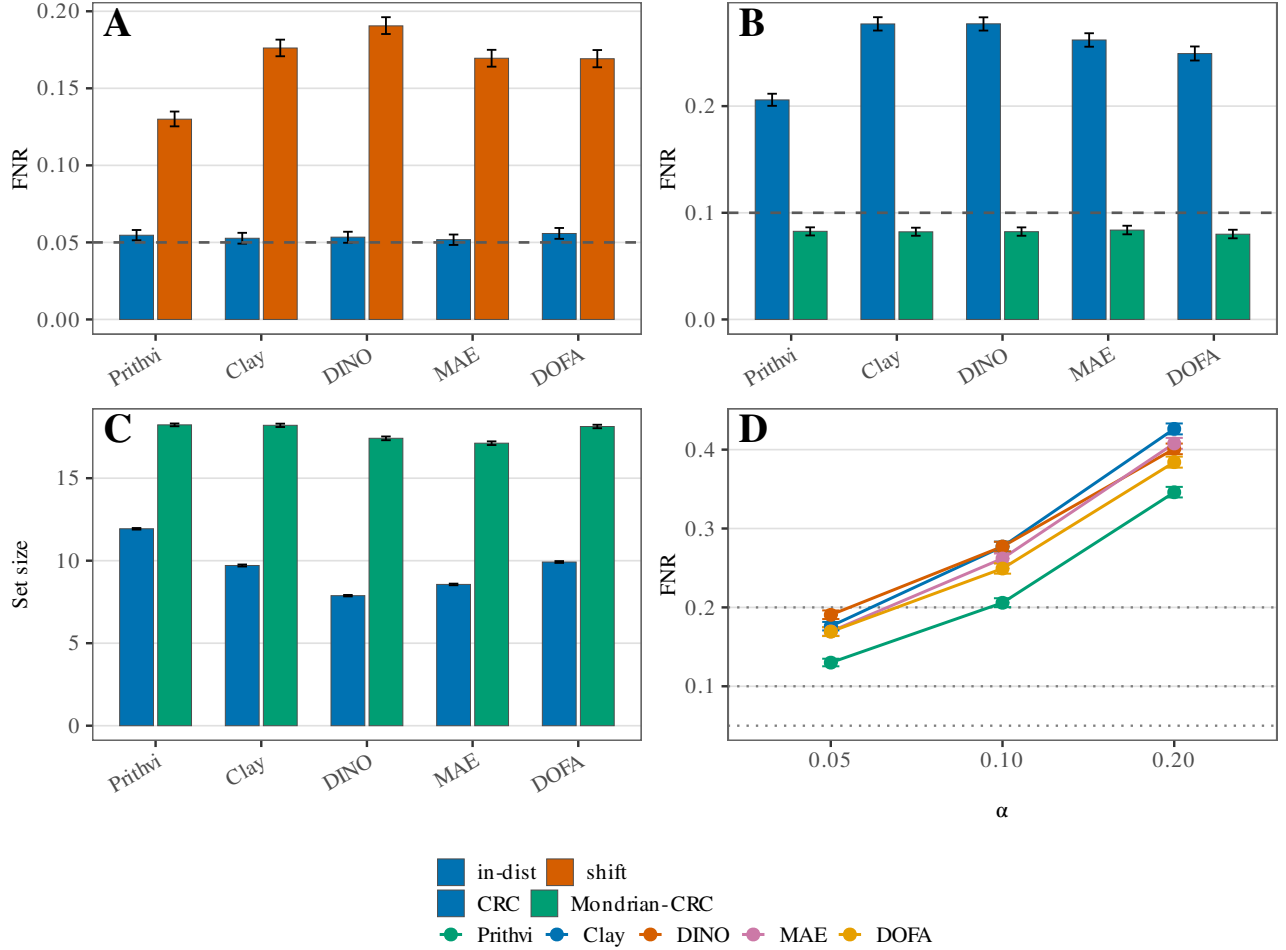


Fig. 8. CRC on BigEarthNet (Table VI; frozen multi-seed bootstrap means with 95% CIs; no new runs). All panels use example-averaged FNR (or set size) pooled over five seeds; error bars are percentile bootstrap intervals ($B=2000$). Encoder ticks and the panel-(d) colour key abbreviate SSL4EO-DINO/MAE as DINO/MAE. (a) Source-calibrated CRC FNR in-distribution vs. under the $E \rightarrow W$ geographic shift at $\alpha=0.05$ (blue = in-dist, vermillion = shift; dashed = nominal α). In-distribution meets the risk level; the same λ incurs large FNR debt under shift. (b) Shift FNR at $\alpha=0.10$: source-calibrated CRC vs. target-side Mondrian-CRC (blue/green; dashed = nominal α). Mondrian-CRC restores FNR to at-or-below nominal for every encoder. (c) Mean set-size cost under shift at the same $\alpha=0.10$ (labels/tile of ~ 27 active): Mondrian-CRC's restore comes with larger sets than source-calibrated CRC (shared method colours with (b)). (d) Source-calibrated CRC FNR under shift across $\alpha \in \{0.05, 0.10, 0.20\}$ by encoder (coloured lines; dotted horizontals = nominal α at each risk level).

F. External validation on a non-European benchmark (GEO-Bench m-so2sat)

Both benchmarks above are European Sentinel-2. To test whether the coverage-debt phenomenon and its target-side repair survive outside Europe, we repeat the harness on GEO-Bench (26) m-so2sat (So2Sat LCZ42 (42); 42 cities across multiple continents; 17 local-climate-zone classes), using GEO-Bench’s own train/valid (source) vs. held-out test-city (target) partition as the shift. Because this release ships no per-tile latitude/longitude in our repo, there is no spatial-Mondrian arm here; we compare only source-only split-conformal against a non-spatial, target-calibrated arm (*target-global*), i.e. exactly the label-access ablation that Table V isolated as the active ingredient on EuroSAT. All five frozen encoders now complete (Prithvi, Clay, SSL4EO ViT-S/16

MAE native-S2, SSL4EO-MAE ViT-B/16, DOFA); the earlier Clay v1.5 checkpoint-load failure (> 4 GB expected in cache) was resolved on the extraction host, so the encoder-ranking diagnostic is now available on this benchmark too (all five extracted on the hash-gated frozen substrate).

The debt-and-repair pattern replicates outside Europe: under the held-out-city shift, accuracy drops by ≈ 0.14 – 0.18 across the five encoders ($0.670 \rightarrow 0.522$ Prithvi up to $0.796 \rightarrow 0.637$ SSL4EO-MAE)—a genuine, non-trivial distribution shift. Source-only split-conformal under-covers at every risk level and every encoder ($\alpha = 0.05$: 0.877 – 0.904 ; $\alpha = 0.20$: 0.632 – 0.685), and the non-spatial target-calibrated arm restores coverage to near-nominal ($\alpha = 0.05$: 0.950 – 0.963), with the paired-seed restoration CI excluding 0 in every one of the fifteen debt cells (5 encoders $\times$ 3 risk levels; Table VII). Restoration is bought at modest set sizes (~ 2 – 6 of 17 classes), not vacuous full-label sets. This external label-access replication now spans all five encoders on a non-European, multi-continent benchmark: it confirms the operative part of the finding—*target-side conformal recalibration*, not spatial binning, repays the debt. It still lacks a spatial-Mondrian arm (no per-tile lat/lon), so it tests the label-access mechanism only; but with all five encoders now loadable it also carries the per-class conditional-coverage diagnostic (below). A data-provenance note applies: m-so2sat was obtained via GEO-Bench 1.0 (HuggingFace recursix/geo-bench-1.0) after the designed v0.9.1 Zenodo pull was unreachable, with a label map reconstructed from per-sample ground truth (21,964/21,964 samples, 0 failures); the v0.9.1 $\rightarrow$ v1.0 content differences were not independently audited (see the data-provenance note in the code archive).

As a second, multiclass conditional-coverage dissociation, with all five encoders loadable, m-so2sat also lets us re-run the per-class conditional-coverage diagnostic of the EuroSAT conditional finding (§III) on a second, 17-class multiclass benchmark. At $\alpha=0.05$ we measure, for the source-only split predictor, the worst individual LCZ class coverage per encoder (Table VIII). It dissociates from both shifted accuracy and shift-ECE: Spearman(worst-class cov, acc) = -0.10 ($p=0.87$, 5 pairwise inversions) and Spearman(worst-class cov, ECE_{sh}) = $+0.60$ ($p=0.29$, 3 inversions); the class-coverage spread scrambles likewise ($\rho = -0.30$ against each, 6 inversions). The most accurate encoder here (SSL4EO-MAE, acc 0.637) is not the one that best protects its worst class, and the worst-covered class (0.689 – 0.733 across encoders, under marginal coverage 0.877 – 0.904) is under-served regardless of accuracy or calibration ranking. This corroborates the EuroSAT conditional-coverage finding with an independent multiclass benchmark—though on the unrepaired split sets here (no target-side repair or spatial arm on this release), and, like the EuroSAT case, it is post-hoc, exploratory, and small- n ($n=5$, all $p > 0.28$; §V).

TABLE VII

NON-EUROPEAN EXTERNAL VALIDATION ON GEO-BENCH $m-s02sat$ (17 LOCAL-CLIMATE-ZONE CLASSES, 42 MULTI-CONTINENT CITIES; 5-SEED MEANS). TARGET COVERAGE “SPLIT / TARGET-GLOBAL” UNDER THE HELD-OUT-CITY SHIFT (SPLIT = SOURCE-ONLY CALIBRATION; TARGET-GLOBAL = NON-SPATIAL, TARGET-CALIBRATED LABEL-ACCESS ARM; NO SPATIAL-MONDRIAN ARM, AS THIS RELEASE CARRIES NO PER-TILE LAT/LON IN OUR REPO). SET SIZES (OF 17 CLASSES) IN BRACKETS. ALL FIVE ENCODERS NOW LOAD (CLAY v1.5 CHECKPOINT-LOAD FAILURE RESOLVED).

α	Encoder	Split [set]	Target-global [set]	Acc. in→shift
0.05	Prithvi	0.904 [3.92]	0.956 [5.72]	0.670→0.522
0.05	Clay	0.891 [3.32]	0.954 [5.08]	0.713→0.569
0.05	SSL4EO ViT-S/16 MAE	0.880 [2.42]	0.962 [4.25]	0.781→0.601
0.05	SSL4EO-MAE ViT-B/16	0.882 [2.18]	0.950 [3.47]	0.796→0.637
0.05	DOFA	0.877 [2.49]	0.963 [5.35]	0.777→0.608
0.10	Prithvi	0.825 [2.71]	0.907 [4.02]	0.670→0.522
0.10	Clay	0.808 [2.27]	0.888 [3.24]	0.713→0.569
0.10	SSL4EO ViT-S/16 MAE	0.775 [1.67]	0.904 [2.75]	0.781→0.601
0.10	SSL4EO-MAE ViT-B/16	0.795 [1.57]	0.896 [2.38]	0.796→0.637
0.10	DOFA	0.781 [1.69]	0.918 [3.30]	0.777→0.608
0.20	Prithvi	0.685 [1.62]	0.819 [2.65]	0.670→0.522
0.20	Clay	0.672 [1.39]	0.795 [2.17]	0.713→0.569
0.20	SSL4EO ViT-S/16 MAE	0.632 [1.08]	0.809 [1.88]	0.781→0.601
0.20	SSL4EO-MAE ViT-B/16	0.654 [1.04]	0.793 [1.59]	0.796→0.637
0.20	DOFA	0.634 [1.07]	0.823 [1.99]	0.777→0.608

TABLE VIII

PER-CLASS CONDITIONAL COVERAGE ON GEO-BENCH $m-s02sat$ (17 LCZ CLASSES; SOURCE-ONLY SPLIT-CONFORMAL, $\alpha=0.05$; 5 SEEDS). THE WORST INDIVIDUAL-CLASS COVERAGE DISSOCIATES FROM SHIFTED ACCURACY AND SHIFT-ECE (Spearman = $-0.10 / +0.60$; 5/3 PAIRWISE INVERSIONS), A SECOND MULTICLASS INSTANCE OF THE EUROSAT CONDITIONAL-COVERAGE COLLAPSE.

Encoder	Acc. shift	ECE shift	Marg. cov	Worst-class cov	Class spread
Prithvi	0.522	0.364	0.904	0.731	0.269
Clay	0.569	0.325	0.891	0.732	0.264
SSL4EO ViT-S/16 MAE	0.601	0.114	0.880	0.689	0.302
SSL4EO-MAE ViT-B/16	0.637	0.173	0.882	0.702	0.269
DOFA	0.608	0.214	0.877	0.733	0.253

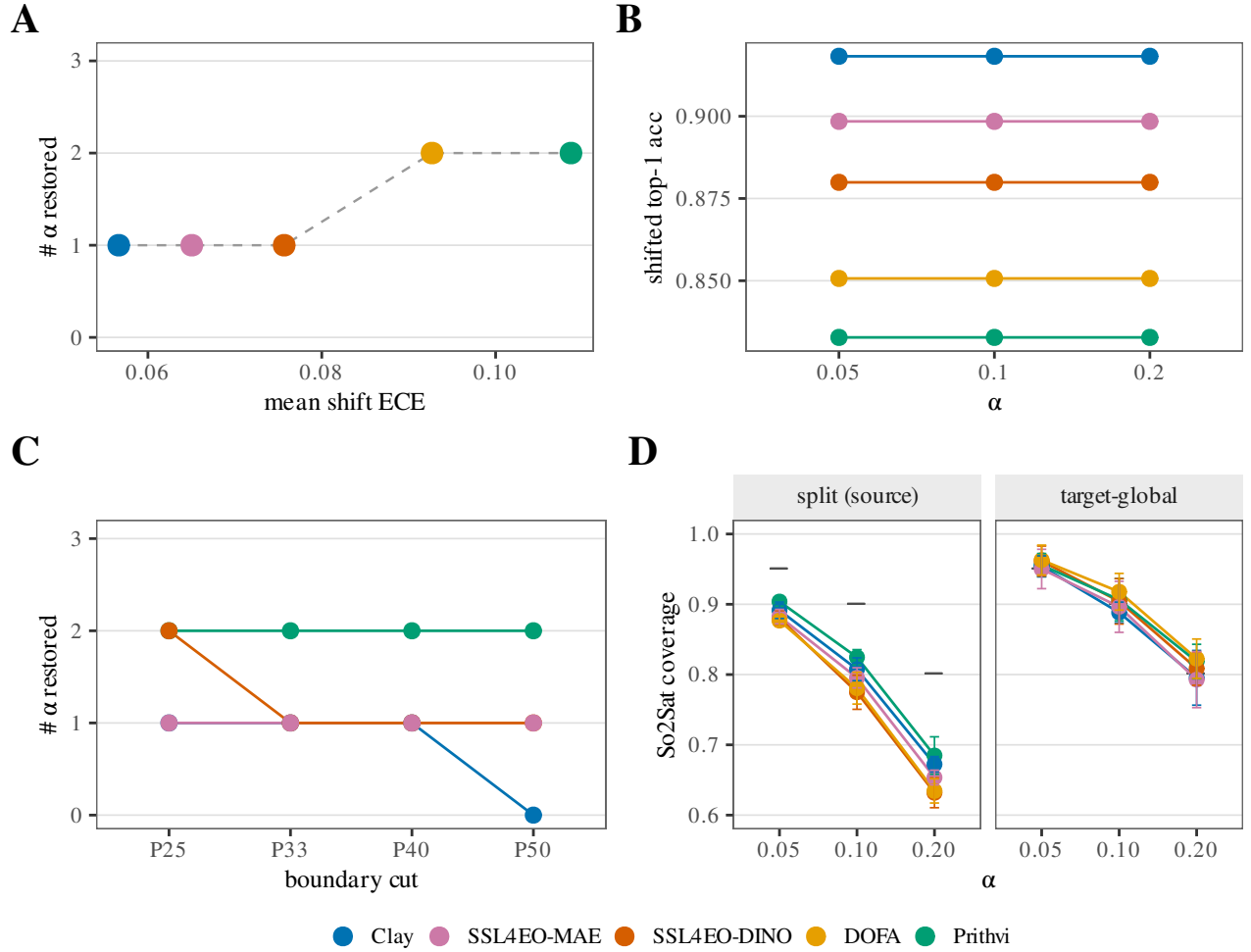
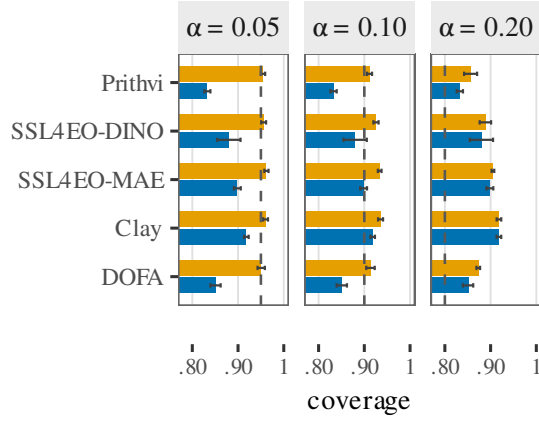
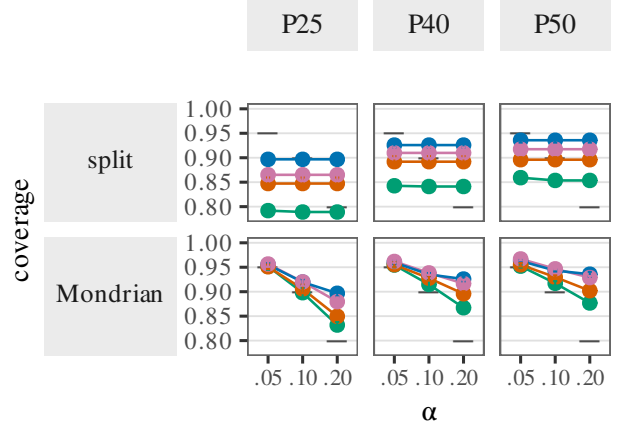
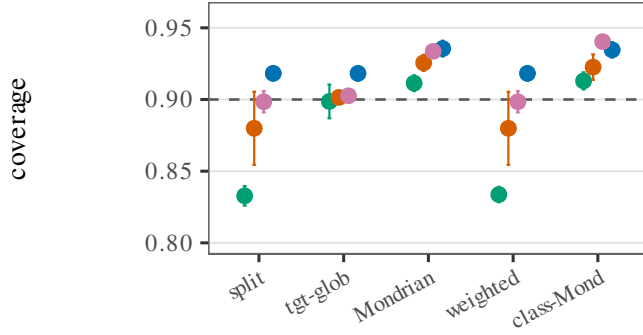
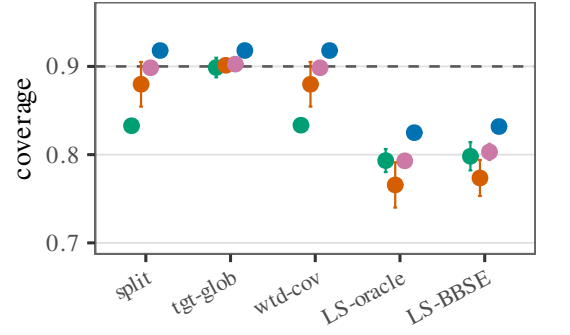


Fig. 9. Four-panel coverage-debt diagnostic across five frozen encoders (shared colour legend: Clay, SSL4EO-MAE, SSL4EO-DINO, DOFA, Prithvi). (A) Mean in-shift ECE (coverage debt) vs. the number of risk levels $\alpha \in \{0.05, 0.10, 0.20\}$ where spatial-Mondrian *restores* target coverage—i.e. the paired gap CI excludes 0 in the positive direction. Debt orders Prithvi > DOFA > SSL4EO-DINO > SSL4EO-MAE > Clay; repair counts are weakly monotone (Prithvi = DOFA 2, others 1). (B) Shifted top-1 accuracy vs. α (flat across risk levels; ranking matches debt). (C) Boundary-sweep check on the four battery encoders (DOFA absent from that frozen sweep): # α restored at spatial cuts P25–P50. (D) So2Sat LCZ cross-continent label-access check: source-split vs. target-global coverage (facets), with grey ticks marking nominal $1 - \alpha$ and error bars seed-level CIs. Set-size cost of repair is in Fig. 11.

A**B****C****D**

■ split (source) ■ spatial-Mondrian ● Prithvi ● SSL4EO-DINO ● SSL4EO-MAE ● Clay

Fig. 10. Coverage restoration under real geographic shift (error bars = seed-level t -CIs, $n=5$; dashed = nominal $1 - \alpha$). (a) EuroSAT P_{33} : source split-conformal vs. spatial-Mondrian coverage by encoder and α (five encoders, including DOFA). (b) Same split vs. Mondrian contrast at non- P_{33} spatial cuts (P25/P40/P50); row = arm, colour = encoder (four-encoder battery; DOFA absent from this frozen record). (c) Five conformal arms at $\alpha=0.10$ (ticks: split, tgt-glob, Mondrian, weighted, class-Mond). (d) Label-shift controls at $\alpha=0.10$ (ticks: split, tgt-glob, wtd-cov, LS-oracle, LS-BBSE). Spatial-Mondrian restores where debt exists; DOFA's shifted split coverage is 0.851 at every α and spatial-Mondrian lifts it to 0.951/0.913/0.873.

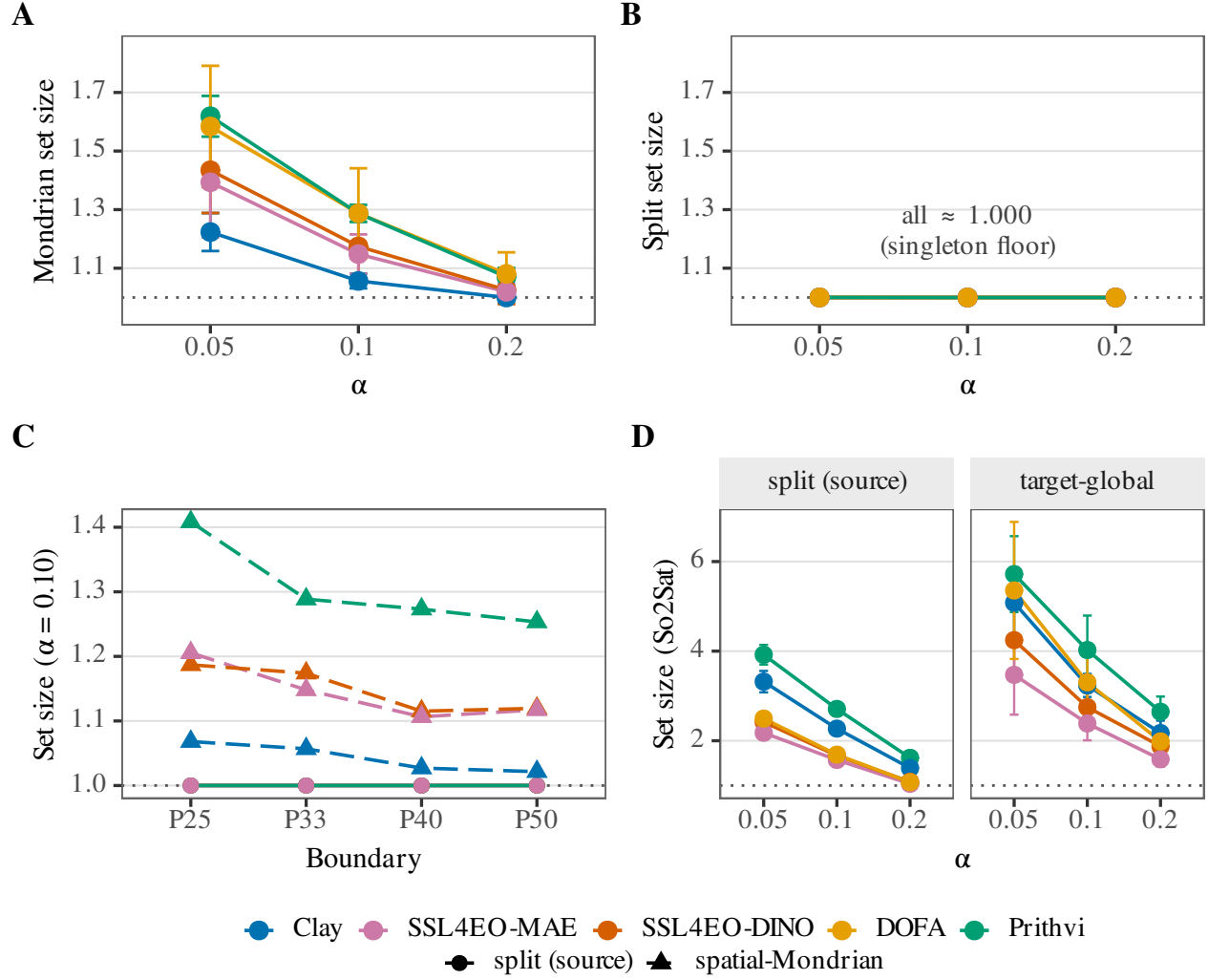


Fig. 11. Four-panel set-size cost of restoration (error bars seed-level t -CIs, $n=5$; dotted line = singleton floor of set size 1). Shared encoder colours throughout; panel (c) uses linetype/shape for split vs. spatial-Mondrian. (a) EuroSAT spatial-Mondrian mean set size vs. α (rises above the floor only where Mondrian repays coverage debt; DOFA 1.58/1.29/1.08 at $\alpha = 0.05/0.10/0.20$). (b) Source split-conformal set size on the *same* y -scale as (a): every encoder sits on the singleton floor (≈ 1.000), so the excess size in (a) is the restoration cost. (c) Set size vs. spatial boundary percentile at $\alpha=0.10$ (solid = source split; dashed = spatial-Mondrian). (d) So2Sat LCZ (17 classes) source-split vs. target-global set size in side-by-side facets—larger absolute sets than EuroSAT, on its own y -scale.

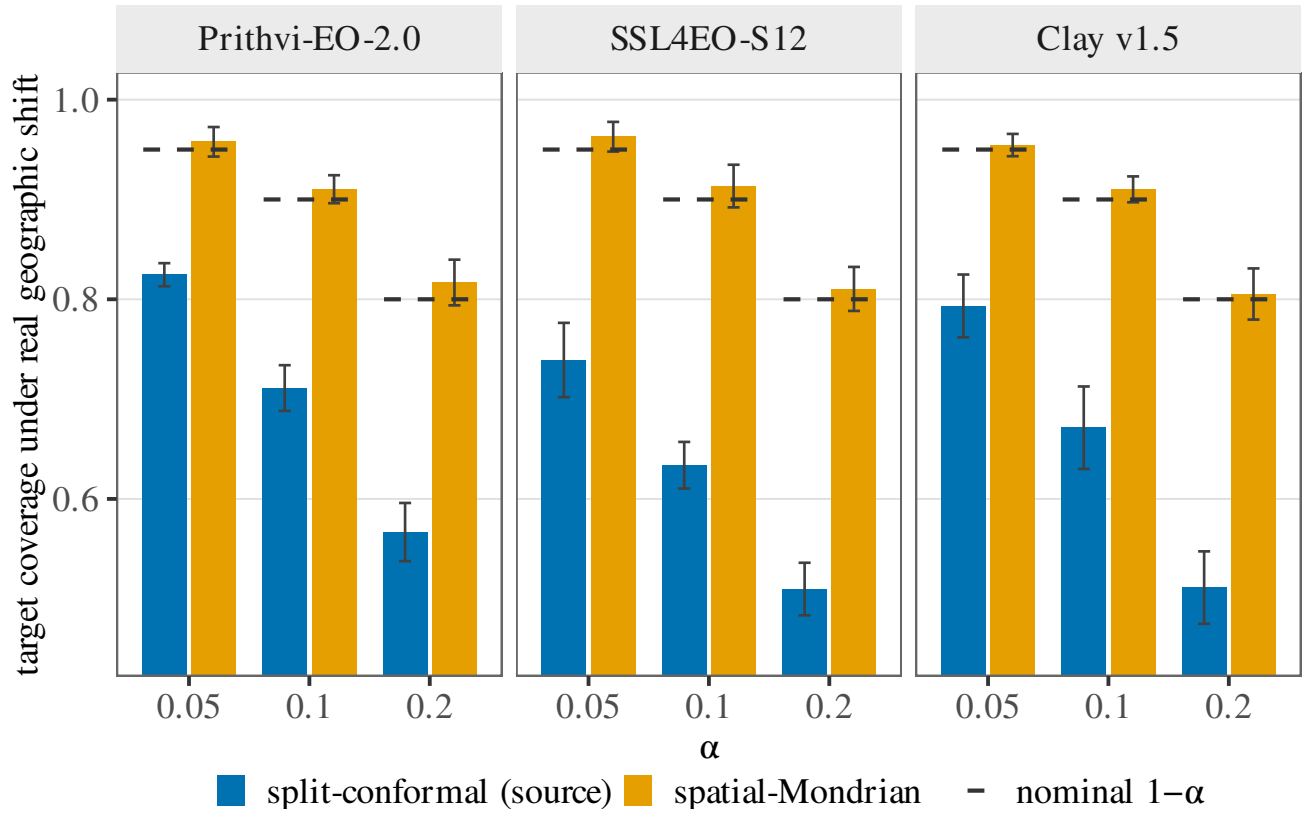


Fig. 12. Dominant-label proxy on GEO-Bench *m-bigearthnet* (single-label reduction; 12 classes): coverage debt and spatial-Mondrian restoration replicate across all three frozen FMs (dashed = nominal $1 - \alpha$; error bars seed-level *t*-CIs, $n=5$). A coarse, qualitative replication; the per-label analysis of Fig. 13 is primary.

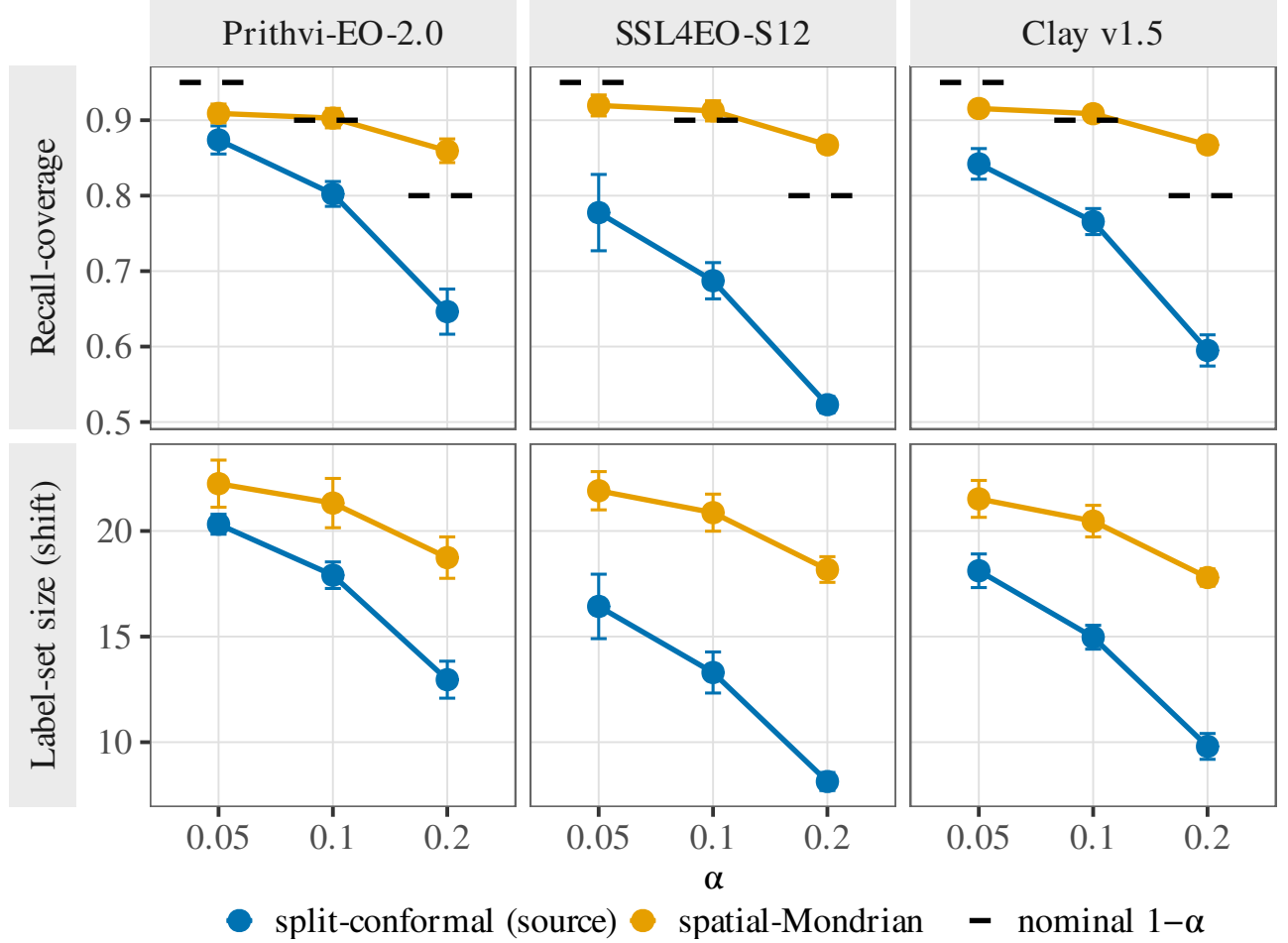


Fig. 13. Per-label multi-label conformal on GEO-Bench (26) m-bigearthnet (BigEarthNet-S2; 27 active labels; real geographic E→W shift; 5 seeds). Top: mean per-label recall-coverage $\Pr(c \in \mathcal{C}(x) \mid y_c = 1)$ for source split-conformal and target-side spatial-Mondrian per FM and α (dash = $1 - \alpha$). Bottom: mean predicted label-set size per tile. Mondrian restores per-label coverage at $\alpha \in \{0.10, 0.20\}$ (paired CI across labels excludes 0) for all three FMs, confirming coverage debt in a rigorous multi-label setting.

IV. DISCUSSION

In plain terms: the audit earns its keep on the FNR-versus-set-size tradeoff, the singleton-degeneracy boundary, and the per-class collapse—and is explicit, rather than silent, about the encoder-ranking and accuracy questions it cannot yet answer with confidence. The through-line is that a conformal audit of a frozen GFM earns its keep exactly where accuracy and calibration go silent: on the multi-label FNR risk-control guarantee, every one of thirteen frozen encoders incurs a geographic FNR debt under shift ($2.6\text{--}4.7\times$ nominal) that only a labeled target slice repairs, and accuracy does not reliably say which encoder is most robust—the least accurate is the most robust, but at $n=13$ the rank correlation is a weak positive ($\rho = +0.32$; Table I)—and on per-class conditional coverage, where restoring the marginal guarantee still leaves one land-cover class covered at 0.850 under a 0.956 marginal (Table II), neither predictable from any scalar a practitioner would rank on. Symmetrically, we are explicit about where the audit does not add information: in the marginal single-label singleton regime the set-size axis is a deterministic restatement of accuracy, so EuroSAT at full data serves as a mapped negative control rather than a decoupling, and the preregistered efficiency-decoupling route is falsified. Within that boundary, the coverage-debt ranking remains a useful descriptive diagnostic. This secondary ranking is a debt-targeted phenomenon whose benefit is governed by the frozen encoder’s calibration robustness, ordering Prithvi, SSL4EO-DINO, SSL4EO-MAE, and Clay consistently across risk levels. This ranking (and the headline coverage-restoration numbers throughout) is reported from an unadjusted logistic probe, i.e. not adjusting for the source/target class-prior mismatch the P_{33} cut induces (Methods; up to $6.54\times$ enrichment / $\sim 4.7\times$ depletion for individual EuroSAT classes; Fig. 14). As a control, we re-fit every probe with class-balanced sample weighting on the identical cached embeddings (Table X): the encoder ranking by shift-ECE, the per-encoder count of risk levels repaired, and the analogous per-label-balanced check on the BigEarthNet per-label conformal analysis are all unchanged to within the seed noise, so the class-prior mismatch does not appear to be driving the ranking—but the unadjusted numbers remain the ones reported as the headline throughout, and the balanced control was run only at the single P_{33} boundary (the boundary sweep of Table V uses the unadjusted probe). The ViT-B backbone (SSL4EO-MAE, 768-dim) incurs less debt than the ViT-S backbone (SSL4EO-DINO, 384-dim); however, these two checkpoints differ simultaneously in architecture size, pretraining objective (MAE vs. DINO), and specific checkpoint (Methods), so this pairwise comparison is consistent with, but does not isolate, a backbone-capacity mechanism for geographic calibration robustness (this pairwise ordering is also unchanged under the balanced-probe control). We deliberately avoid a “universal fix” claim: where an encoder is robust enough that shifted accuracy meets the nominal level, there is no debt and no repair.

A. Class-prior shift and a balanced-probe control

TABLE IX
EUROSAT-MS PER-CLASS FREQUENCY AT THE P_{33} LONGITUDE CUT (SOURCE = E/CENTRAL EUROPE, $n=18,090$; TARGET = W/IBERIAN-ATLANTIC EUROPE, $n=8,910$). CLASSES NOT LISTED SHIFT BY $<2.4\times$.

Class	Source %	Target %	Target/Source ratio
5 (Pasture)	2.62	17.13	$6.54\times$ (enriched)
2 (HerbaceousVegetation)	7.68	18.07	$2.35\times$ (enriched)
9 (SeaLake)	14.10	5.04	$0.36\times$ (depleted)
8 (River)	11.60	4.51	$0.39\times$ (depleted)
1 (Forest)	15.02	3.16	$0.21\times$ ($\sim 4.7\times$ depleted)

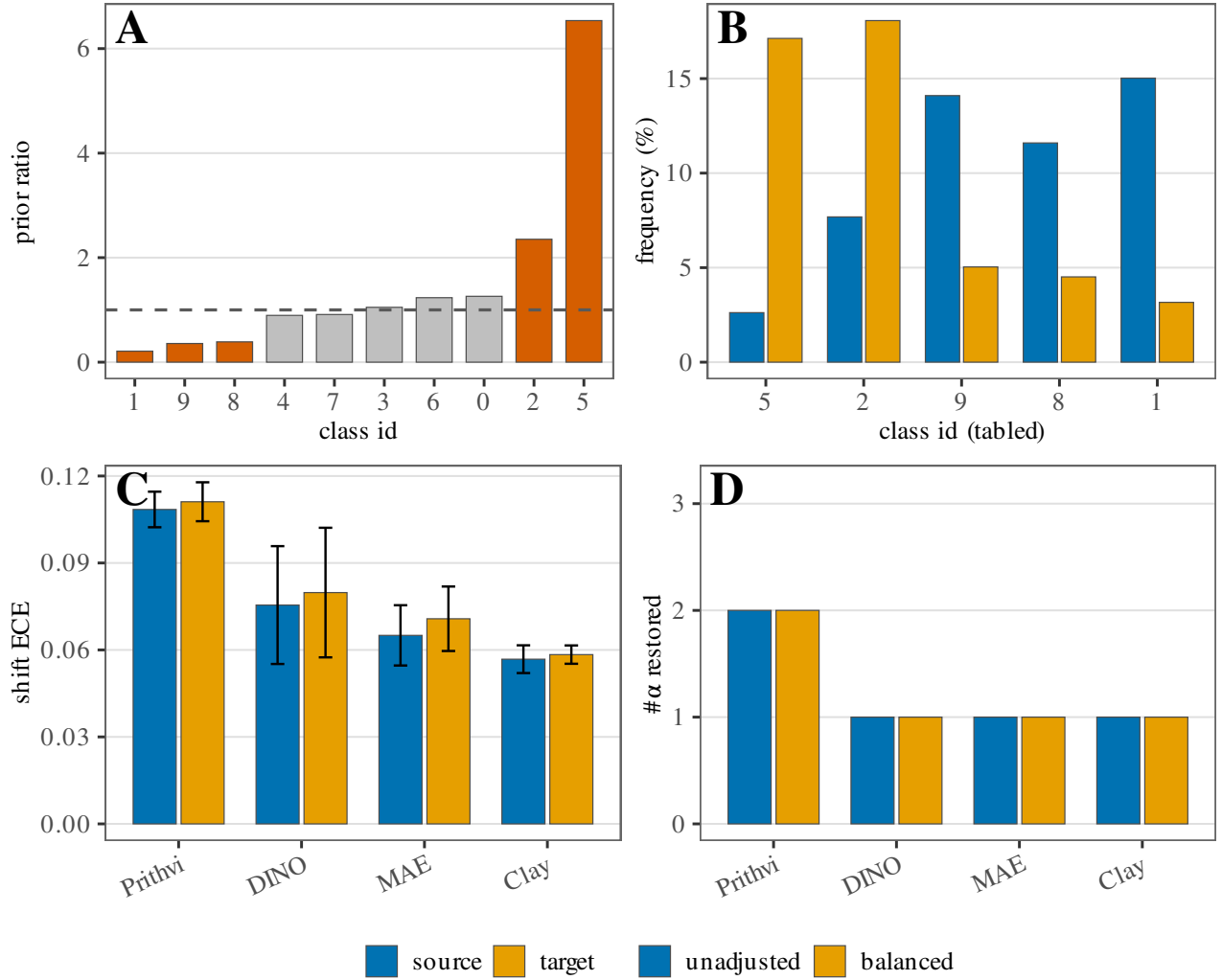


Fig. 14. Four-panel class-prior / balanced-probe control from the frozen P_{33} recompute (Tables IX, X; no new runs). (a) Target/source prior ratio by EuroSAT class. (b) Source vs. target class frequency (%). (c) Unbalanced-probe shift ECE at $\alpha=0.05$. (d) Mondrian coverage under unbalanced vs. balanced probes (where the balanced arm exists).

The encoder ordering by shift-ECE (Prithvi > SSL4EO-DINO > SSL4EO-MAE > Clay) and the number of risk levels each encoder's spatial-Mondrian arm repairs are identical under the balanced control. The analogous per-label-balanced refit of the BigEarthNet per-label conformal probes (same cached embeddings, $\alpha = 0.10$) likewise leaves mean per-label debt and restoration essentially unchanged (e.g. Prithvi debt $0.117 \rightarrow 0.113$, SSL4EO $0.239 \rightarrow 0.236$, Clay $0.163 \rightarrow 0.161$; restoration CI still excludes 0 for all three FMs). Full per-seed numbers, the per-class frequency table for all 10 classes, and the BigEarthNet balanced-control results are included in the code archive.

B. A deployment report card

An illustration of the audit's operational use (post-hoc; not a preregistered claim). Concretely, suppose an agency must monitor multi-label land cover in an unseen region with a frozen off-the-shelf GFM at risk level $\alpha=0.05$, and picks its encoder the way a practitioner would—on accuracy. The median-accuracy encoder (Clay v1.5; in-distribution mAP 0.502, rank 3 of 5) is a deliberately unremarkable choice. Table XI is the reliability report card the audit returns for it, assembled entirely from the frozen result records (no new computation). Deployed uncalibrated, its example-averaged FNR guarantee is missed by $\approx 3.5\times$ (0.176 vs. the promised 0.05); a 30% labeled target-region calibration slice with target-side Mondrian CRC restores it to 0.022, at the cost of larger prediction sets (13.3 $\rightarrow$ 23.8 labels of 27). A second, non-European region (GEO-Bench *m-so2sat*, LCZ-17) shows the complementary caution: a healthy marginal coverage can still hide a land-cover class covered at only 0.732. The recommended operating point is therefore to deploy this encoder only with a labeled target-region recalibration slice, and to monitor per-class coverage, not merely the marginal.

V. LIMITATIONS

In plain terms: our one preregistered confirmatory result is a falsification, our cross-encoder accuracy claim is weaker than a five-encoder pilot suggested, and our geographic reach is Europe plus one non-European benchmark. This section is a single continuous preregistration-disclosure narrative, not a conventional caveats list, and it is deliberately long because it is by design as load-bearing as the results themselves. It discloses, in order: (i) the paper's one preregistered outcome and how the encoder-dependent claim reported in the main text reframes it (the Stage-2 *universal-restoration* gate); (ii) the power limits of every post-hoc decoupling analysis added after that gate, including the roster-expansion result that weakened rather than strengthened, and the small- n conditional-coverage findings; and (iii) the audit's scope boundaries—single-sensor, single-continent until the So2Sat extension, and encoder-roster size—together with the negative-control battery run to rule out confounds. A first-time reader should treat what follows as the paper's own accounting of what its evidence does and does not license, not as an appendix of afterthoughts.

Before the specific disclosures, it is worth stating the integrity checks that ran on every retained cell and passed,

since two of them are what caught the anomalies below in the first place rather than letting them reach a reported number. Every encoder's frozen feature cache is gated at extraction by shape and finiteness assertions (expected embedding width, all-finite, non-degenerate variance); a smoke gate applied this check on a small tile slice for each of the eight roster-expansion encoders before committing the full 7,669-tile run, and the two arms that failed it (*dofa_large*, *prithvi_600m*) were dropped rather than scored (§II). Every retained encoder's in-distribution probe accuracy is checked against a chance-multiplier threshold (a guard added after an earlier incident in which a dead in-distribution control silently produced a spurious coverage-restoration result); this is the same guard that caught the withheld cross-sensor collapse below rather than letting it stand as a reported finding. Every result record additionally carries a code-provenance hash, and feature caches moved between hosts are byte-verified before use. None of this substitutes for the disclosures that follow—it is why they were caught.

As a preregistration disclosure and exploratory reframe: the Stage-2 analysis was preregistered with a *universal-restoration* success criterion: spatial-Mondrian restoration at ≥ 2 of 3 risk levels for both Prithvi and Clay (H3–H4). That criterion was not met, because Clay was restored at only 1 of 3 levels (Prithvi met it with 2/3). The conditional, encoder-dependent claim reported throughout this paper is therefore an exploratory reframe of that preregistered outcome—the debt-targeted framing was adopted after observing that repair occurs exactly where debt exists—and should be read as hypothesis-generating rather than confirmatory; the original criterion outcome is preserved in the frozen analysis record. In confirmatory terms, the single preregistered result this paper delivers is the falsification of the universal-restoration hypothesis; the encoder-dependent account built on top of it is hypothesis-generating. A distinct firm positive stands beside that negative: the within-encoder FNR-debt-and-repair pattern is not itself a preregistered population claim, but a distribution-free sign test computed post-hoc over the thirteen frozen per-encoder estimates confirms it at $p \approx 2.4 \times 10^{-4}$ in both the debt and the repair direction (§III). In scope terms, this paper introduces no new estimator: we reuse split and spatial-Mondrian conformal prediction (22, 33) and Angelopoulos–Bates conformal risk control (24) throughout, and the operative axis is access to a small labeled target-region calibration slice, not a new method—the contribution is the reusable pre-deployment report-card protocol (§IV-B) and the audited evidence behind it, not conformal machinery.

The re-centered decoupling axes are post-hoc and (for the conditional analyses) small- n : the three analyses—the multi-label FNR decoupling, the per-class conditional-coverage collapse, and the singleton-degeneracy boundary—were all run after the preregistered gate, on the same frozen-feature caches, and are exploratory. Two power caveats bound them. First, the cross-encoder FNR comparison is the one claim the roster expansion actively weakened, and we report the weakened result. Over the five-encoder battery the shifted FNR rank-correlated with mAP at $+0.90$ (asymptotic $p=0.037$, though its *exact* permutation p was already 0.083); expanding to the

TABLE X

CLASS-PRIOR-SHIFT CONTROL: CLASS-BALANCED SAMPLE WEIGHTING VS. THE PAPER’S UNADJUSTED LOGISTIC PROBE, RE-FIT ON THE IDENTICAL CACHED FROZEN EMBEDDINGS (NO GPU RE-EXTRACTION), 5 SEEDS, REAL $E \rightarrow W$ SHIFT. TOP PANEL (EUROSAT, $\alpha = 0.05$): ECE-SHIFT RANKING AND PER-ENCODER REPAIR COUNT (OF 3 α ’S) ARE UNCHANGED UNDER THE BALANCED CONTROL. BOTTOM PANEL (BIGEARTHNET-S2, $\alpha = 0.10$, PER-LABEL CONFORMAL): THE ANALOGOUS BALANCED-PROBE REFIT OF THE PER-LABEL HARNESS (PREVIOUSLY REPORTED ONLY IN PROSE) LEAVES MEAN PER-LABEL DEBT AND MONDRIAN RESTORATION ESSENTIALLY UNCHANGED FOR ALL THREE AVAILABLE FMS (SSL4EO-MAE WAS NOT RUN IN THE PER-LABEL BIGEARTHNET HARNESS).

Encoder	Unadjusted (paper headline)			Balanced control		
	Acc. shift	ECE shift	# α repaired	Acc. shift	ECE shift	# α repaired
Prithvi	0.833	0.1085	2/3	0.835	0.1111	2/3
SSL4EO-DINO	0.880	0.0755	1/3	0.878	0.0798	1/3
SSL4EO-MAE	0.898	0.0650	1/3	0.895	0.0708	1/3
Clay	0.918	0.0568	1/3	0.917	0.0584	1/3

Encoder	Unadjusted (paper headline)		Balanced control	
	Mean per-label debt	Mondrian restores	Mean per-label debt	Mondrian restores
Prithvi	0.117	Yes	0.113	Yes
SSL4EO-DINO	0.239	Yes	0.236	Yes
Clay	0.163	Yes	0.161	Yes

TABLE XI

DEPLOYMENT RELIABILITY REPORT CARD (ILLUSTRATION; POST-HOC) FOR A MEDIAN-ACCURACY FROZEN ENCODER (CLAY V1.5) MONITORING LAND COVER IN AN UNSEEN REGION AT $\alpha=0.05$. MULTI-LABEL FNR ROWS: GEO-BENCH $m\text{-bigearthnet}$, $E \rightarrow W$ SHIFT; WORST-CLASS ROW: GEO-BENCH $m\text{-so2sat}$ LCZ-17 (SPLIT ARM). BRACKETS ARE SEED-LEVEL t -CIS ($n=5$). ALL VALUES TRACE TO THE FROZEN RESULT RECORDS DESCRIBED IN THE CODE ARCHIVE.

Report-card question ($\alpha=0.05$, encoder Clay)	Value
Multi-label FNR if deployed uncalibrated (relative to the 0.05 target)	0.175 [0.143, 0.208] $\approx 3.5\times$ over
FNR after a 30% target slice (Mondrian CRC)	0.022 [0.015, 0.029]
Efficiency cost (mean set size, of 27 labels)	13.3 $\rightarrow$ 23.8
Worst class, 2nd region ($m\text{-so2sat}$ LCZ) (under a marginal coverage of)	0.732 [0.692, 0.772] 0.891

full $n=13$ roster regresses the statistic to $+0.32$ (permutation $p=0.28$), with 48/78 (62%) pairwise inversions. The strong five-encoder inversion therefore does *not* generalize—we keep only the weaker, directional claim that accuracy is not a reliable guide to FNR robustness: the correlation stays positive (it never reverses to “better encoder $\rightarrow$ better reliability”) and the extreme case survives (Prithvi, least accurate, is the most FNR-robust; the near-calibration-matched Prithvi/Clay pair still dissociates, ECE 0.004 apart yet 0.046 apart in FNR and in the wrong direction), but at $n=13$ this is directional evidence, not a proven inversion. The one companion correlation that survives is the FNR-vs-set-size tradeoff ($\rho = -0.66$, $p=0.017$ at $\alpha=0.10$): robustness is bought with larger sets. Second, the conditional-coverage dissociation is over $n=5$ encoders with all $p > 0.28$ and a rank correlation that flips sign across α ; it rests on the SSL4EO-DINO datum (one class at 0.850 under a 0.956 marginal, the same class in all five seeds, per-seed range 0.826–0.871, $2\times$ the next-worst) plus 6/10 inversions, so the honest claim is “no detectable/stable dependence,” not proven independence. Both are strong concrete decouplings that a practitioner cannot anticipate from accuracy or ECE, and both would benefit from a larger encoder roster—a scoping choice (frozen-feature extraction only), not a capability gap. The singleton-degeneracy boundary, by contrast, is a deterministic property we simply map, not a correlation.

The headline shifts are cross-region (single-sensor Sentinel-2), not cross-sensor. BigEarthNet confirms the phenomenon under both the dominant-label proxy and rigorous per-label conformal; the two analyses agree qualitatively and the per-label treatment is now primary. A preliminary cross-sensor probe (Sentinel-2 calibration transferred to a native Sentinel-1 encoder on paired $m\text{-so2sat}$ tiles) is withheld because its in-distribution control failed the probe-sanity guard described above (accuracy at chance, 0.06, versus 0.513 for the same encoder on the identical subset in a companion run)—an undiagnosed run-specific failure, not a sensor effect, documented in full in the code archive (debug record archived in the code-and-data deposit). We make no cross-sensor claim here and leave a contrastively-aligned S1/S2 encoder, the right vehicle for that question, to future work. Disentangling backbone size from pretraining objective would require a ViT-S-MAE and a ViT-B-DINO checkpoint, a 2×2 cell only partially available (the public SSL4EO-S12 release contains no ViT-B/16+DINO Sentinel-2 checkpoint); we leave it to future work. On encoder diversity, a fifth distinct architectural family (DOFA) has now been added to the headline EuroSAT grid (§III), where it carries the second-heaviest coverage debt of the five—between Prithvi and SSL4EO-DINO—and is restored at 2/3 risk levels, consistent with the encoder ordering reported here. (Its GEO-Bench $m\text{-brick-kiln}$ cell used the same pre-correction

feature path and is withheld pending re-computation; that task's v1.0 data was not available on the correction host.) The encoder roster is illustrative of the diagnostic, not a leaderboard: adding a further 2025-generation family such as Galileo/TerraFM needs only frozen-feature extraction—no fine-tuning—and is a scoping choice rather than a capability gap. On external geographic generalization, the single-continent limitation is now partly addressed: the non-European *m-so2sat* validation above (42 cities, multiple continents) reproduces the debt-and-repair pattern across all 5/5 encoders and 3/3 risk levels (Table VII; the earlier Clay checkpoint-load block is resolved), and now supplies a second multi-class conditional-coverage dissociation (Table VIII), though it remains a label-access ablation without a spatial-Mondrian arm. EuroSAT-MS and the BigEarthNet-S2 extension (36.96–67.80°N) remain European Sentinel-2, so continent-scale generalization of the spatial-Mondrian arm specifically is still future work. The EuroSAT-MS headline numbers were fixed at a single longitude cut (P_{33}) before any alternative was tried; the boundary-sensitivity sweep of Table V is post-hoc (run after the freeze on the cached embeddings) and confirms the encoder ordering and debt-targeted restoration at $P_{25}/P_{40}/P_{50}$, so the $n=1$ -boundary caveat is now resolved empirically, though the headline numbers themselves remain those of the pre-committed P_{33} cut. Finally, the label-access ablation shows that on this benchmark the non-spatial target-calibrated baseline matches spatial-Mondrian, so our results do not demonstrate a benefit of spatial conditioning per se. Two adjusted-conformal baselines a hostile reading would demand have now been run as negative controls: covariate-shift-weighted conformal (Table IV) under-covers for every encoder, and label-shift-adjusted conformal (oracle and BBSE, Table IV) fails to restore coverage and worsens it for three of four encoders—ruling out the class-prior-shift confound the P_{33} cut could otherwise raise. Class-conditional Mondrian conformal remains unexecuted. The target-recalibration planning curve (Fig. 7) is post-hoc and post-freeze exploratory: the frozen protocol hardcodes the calibration budget at 30% of the target region; the 0.5%–30% sweep was run after the freeze on cached embeddings to turn that single point into a practitioner-facing planning curve, reported as a secondary descriptive finding on the same footing as the boundary-sensitivity sweep and the label-access ablation. Its EuroSAT-MS arm reproduces the frozen 30% operating point exactly (zero difference in coverage and set size across all 4 encoders $\times$ 3 risk levels $\times$ 5 seeds). Its BigEarthNet-S2 arm does *not*: the budget-independent source-calibrated CRC FNR agrees with the frozen record to 1.28×10^{-3} against a declared 10^{-3} tolerance, so that gate fails in one of ten cells (float32 vs. float64 residual; reported as failed rather than widened post hoc). The BigEarthNet arm is additionally Prithvi-only.

DATA AND CODE AVAILABILITY

EuroSAT-MS (public, Sentinel-2); frozen weights Prithvi-EO-2.0-300M, Clay v1.5, SSL4EO-S12 DINO (torchgeo), SSL4EO-S12 MAE (wangyi111/SSL4EO-S12, Hugging-Face), all public/no-auth. The code and frozen result records

used to produce every result and figure in this manuscript are available at GitHub (<https://github.com/PeterPonyu/geospatial-fm-reliability-research>; repository handle *PeterPonyu*) and archived at Zenodo, DOI 10.5281/zenodo.21130299 (this DOI is currently reserved on a draft deposition and becomes active when the record is published); reproduction instructions and a fast regression gate are provided in the code archive. Per IEEE's author-reproducibility guidelines, code archiving with a persistent identifier via GitHub plus a DOI-issuing repository (Zenodo) is sufficient for TGRS; IEEE's optional Code Ocean partnership was not used for this submission.

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